COL718
Assignment 1 Report

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Introduction

"The gem5 simulator is a modular platform for computer-system architecture research, encompassing system-level architecture as well as processor microarchitecture. gem5 is a community-led project with an open governance model. gem5 was originally conceived for computer architecture research in academia, but it has grown to be used in computer system design by academia, industry for research, and in teaching." [1]

gem5 allows the simulation of code or a full system using different CPU types, Memory Models, etc. CPUs can be simulated to run at various frequencies.

Performance of the simulated code is evaluated based on the following two performance metrics:

1) **CPI** (Cycles per Instruction): The average number of CPU cycles taken by an instruction to complete execution. A lower CPI implies faster CPU performance. Architectural enhancements like CPU pipelining and out-of-order execution can decrease(improve) a processor's CPI.
2) **Execution Time**: The total time taken by the code to complete in the simulated system in gem5.

Benchmarking

The simulation is set up according to the included config.py file. The following types of components are used to simulate the Matrix Multiplication code.

1) **CPU Types**:
    a) ATOMIC: A type of SIMPLECPU that uses atomic memory access and does in-order processing.
    b) TIMING: A type of SIMPLECPU that simulates memory access and does in-order processing.
    c) MINOR: An in-order processor with 4-stage pipelining.
    d) O3: AN out-of-order processor with 5-stage pipelining.

2) **CPU Frequencies**: 600MHz, 800MHz, 1GHz, 1.2GHz, 1.4GHz, 1.6GHz, 1.8GHz, 2GHz, 2.2GHz, 2.4GHz, 2.6GHz, 2.8GHz, 3GHz, 3.2GHz, 3.3GHz

3) **Memory Models**: SINGLE_CHANNEL_DDR3_1600, SINGLE_CHANNEL_DDR3_2133, SINGLE_CHANNEL_LPDDR3_1600, SINGLE_CHANNEL_DDR4_2400, SINGLE_CHANNEL_DIMM_DDR5_4400, SINGLE_CHANNEL_DIMM_DDR5_6400, SINGLE_CHANNEL_DIMM_DDR5_8400, DUAL_CHANNEL_DDR3_1600, DUAL_CHANNEL_DDR3_2133, DUAL_CHANNEL_LPDDR3_1600, DUAL_CHANNEL_DDR4_2400

A total of 660(4 CPUs x 15 Freq X 11 Memory Models) measurements were taken with gem5 for the X86 ISA. The relevant benchmarking data table is present in Annexure A.2 at the end of this report.

Implementation of Matrix Multiplication(non-trivial)

A cache-optimized(L1d cache) Matrix Multiplication code is used to run the simulations. The relevant code is present in Annexure A.1.

The trivial matrix multiplication (C=AxB) is not cache optimized as each element is accessed in Column-major for for matrix B, which causes cache misses. In order to optimize this, a transpose of matrix B is taken before multiplication, so that the elements in Matrix B are always accessed in Row-major form, thus reducing Cache Misses and overall execution time for large matrix sizes. The size of the matrices used in the simulated code is 100 x 100.

Graph plots and Observations from Benchmarking data

The following is a series of line graph plots obtained for performance metrics(simSeconds and CPI), where one out of three variables(CPUType, CPUFrequency, and MemoryModel) is fixed, and the remaining two variables are used to plot simSeconds and CPI.

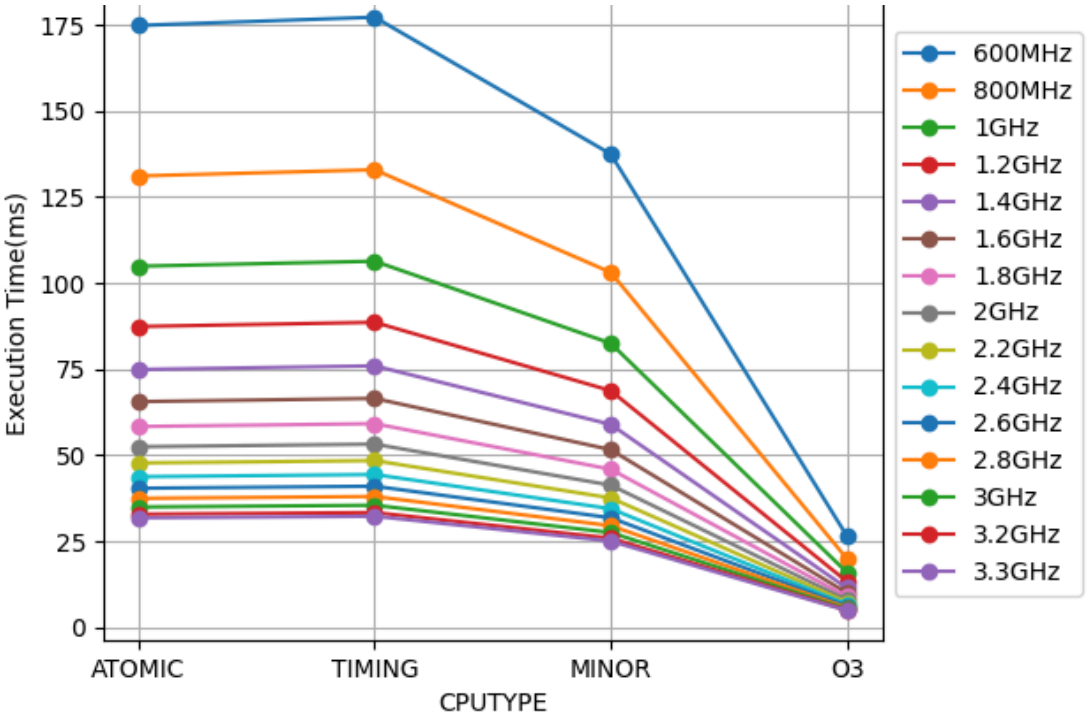


Figure 1: CPUTYPE vs SIMSECONDS for different CPUFREQUENCY(SINGLE_CHANNEL_DDR3_1600).

- In Figure 1, the Execution time(simSeconds) decreases as the CPU frequency is increased, and the Memory model(SINGLE_CHANNEL_DDR3_1600) is kept constant.

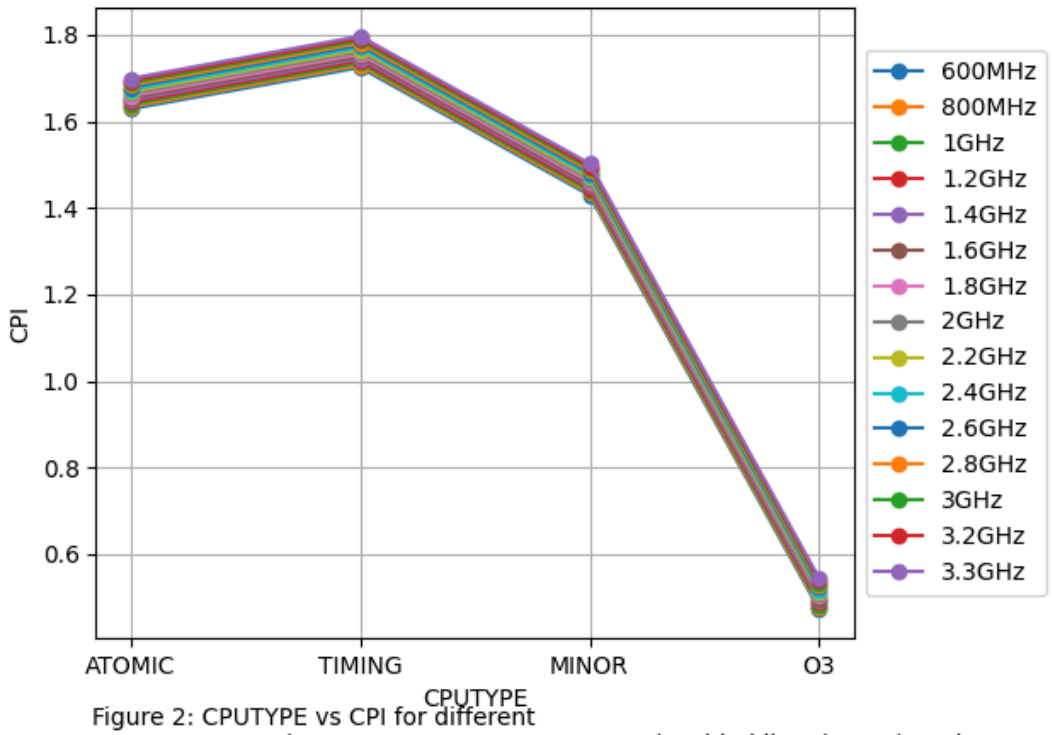


Figure 2: CPUTYPE vs CPI for different CPUFREQUENCY(SINGLE_CHANNEL_DDR3_1600). Added jitter(0.005) to show overlapping line graphs.

- In Figures 1 and 2, while keeping the memory model constant, the TIMING CPU has the worst performance, as it simulates memory access time but offers no architectural optimizations. The Atomic CPU has better performance than CPU as it does not simulate memory access correctly. MINOR CPU has better performance(less CPI, less Execution Time/simSeconds) over TIMING and ATOMIC, as it processes instructions using a 4-stage pipeline. Finally, the O3CPU has the best performance as it offers a 5-stage pipeline and out-of-order execution. This same behavior is observed for all 15 CPU frequencies tested.

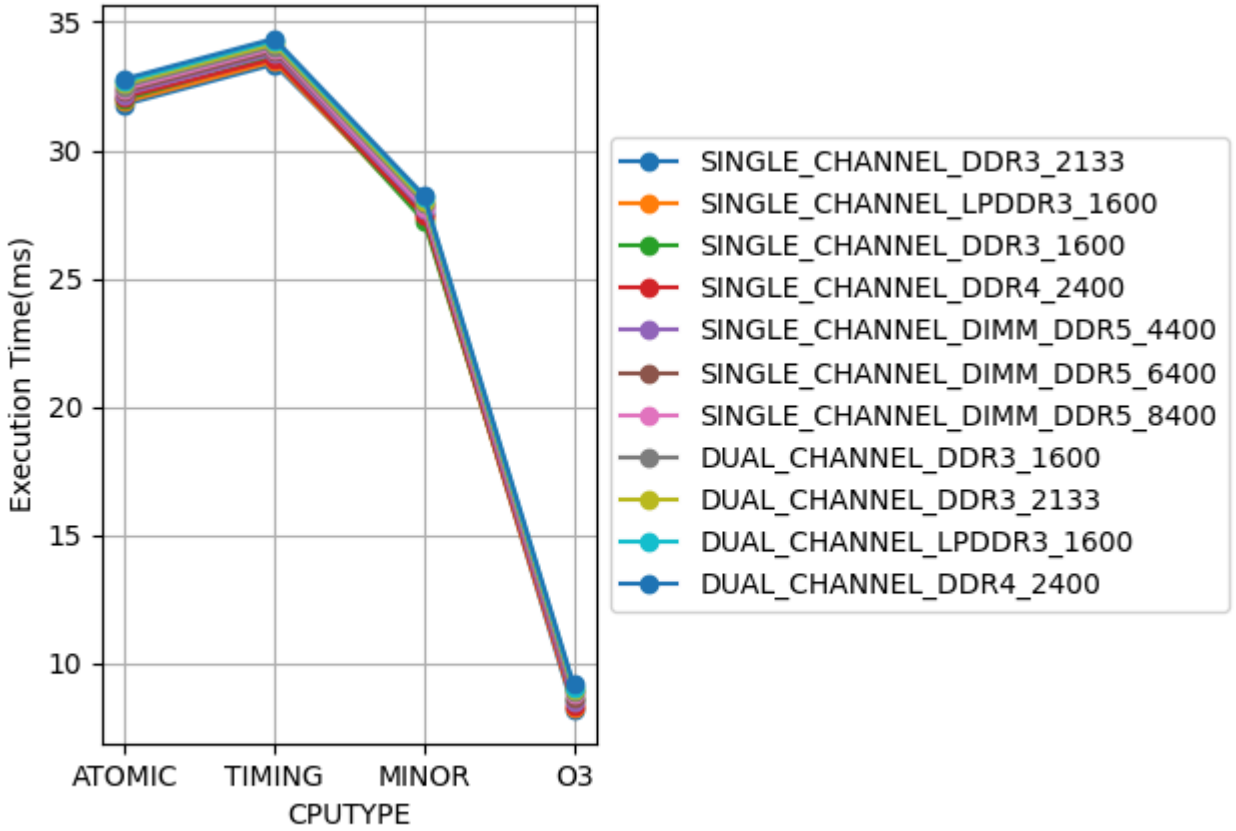


Figure 3: CPUTYPE vs SIMSECONDS for different MEMORYMODEL(3.3GHz). Added jitter(0.1) to show overlapping line graphs.

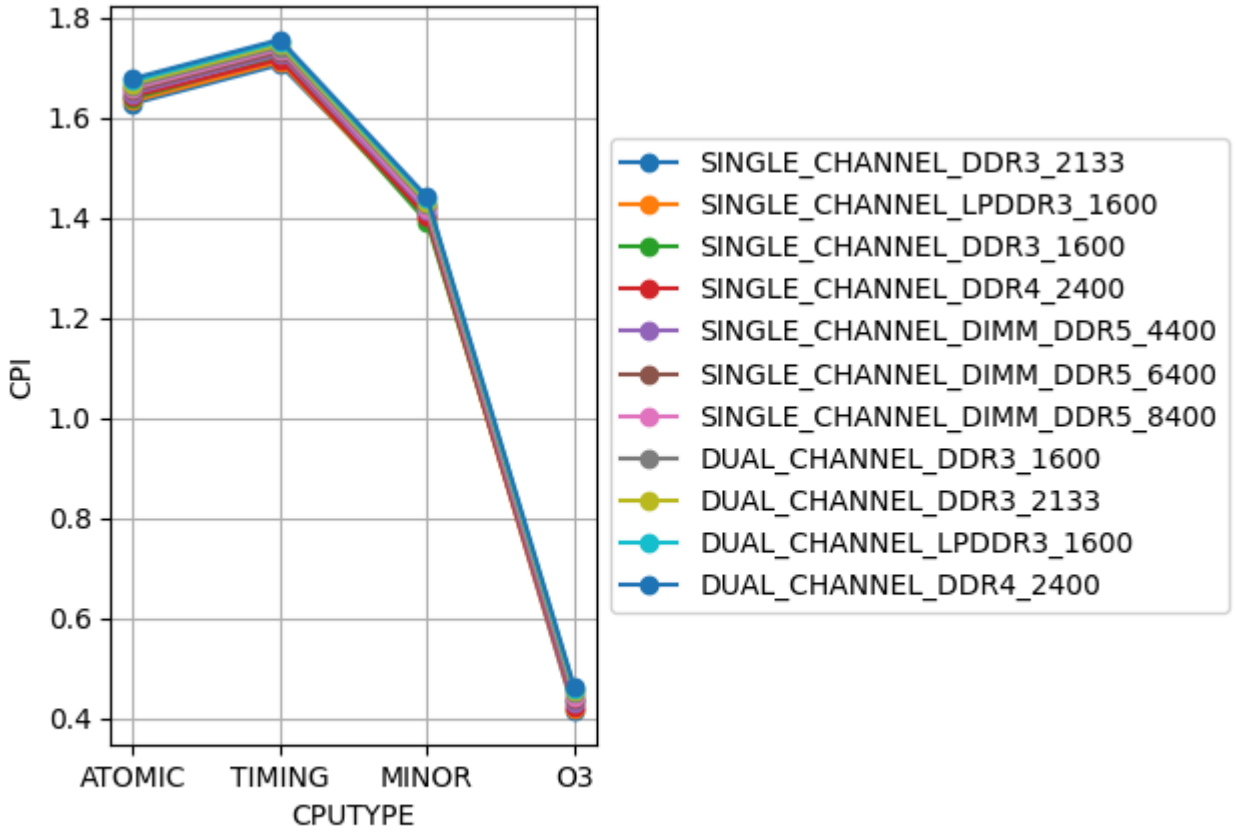


Figure 4: CPUTYPE vs CPI for different MEMORYMODEL(3.3GHz). Added jitter(0.005) to show overlapping line graphs.

- Figures 3 and 4 show the same performance behavior as Figures 1 and 2 for different Memory Models, while keeping a CPU frequency of 3.3GHz constant.

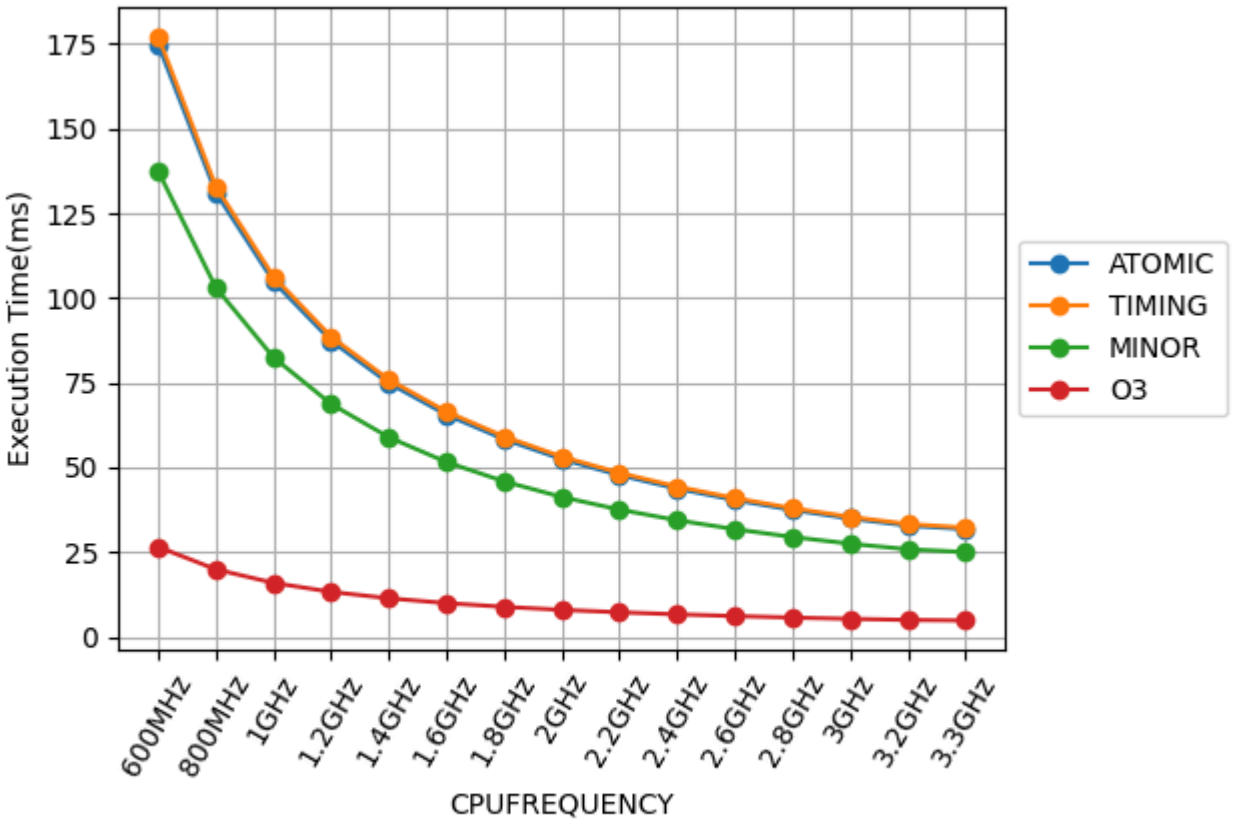


Figure 5: CPUFREQUENCY vs SIMSECONDS for different CPUTYPE(SINGLE_CHANNEL_DDR3_1600).

- In Figure 5, for all four types of CPUs, execution time(simSeconds) decreases as CPU frequency is increased.

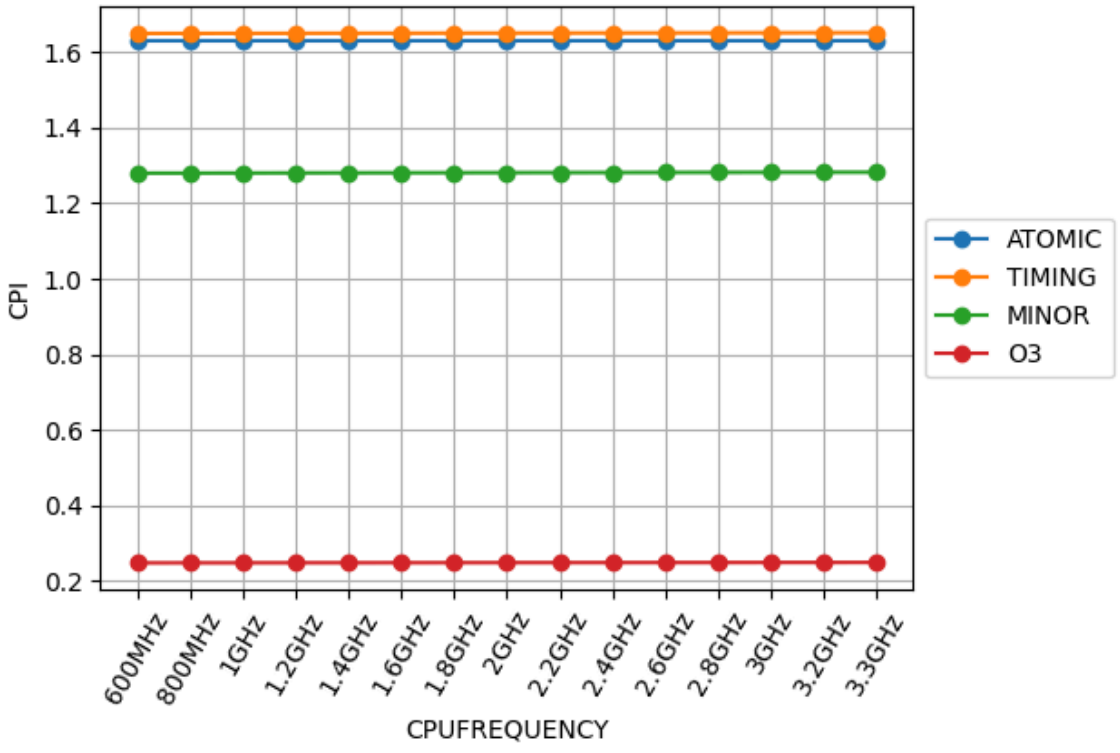


Figure 6: CPUFREQUENCY vs CPI for different CPUTYPE(SINGLE_CHANNEL_DDR3_1600).

- In Figure 6, for all four types of CPUs, CPI remains the same as CPU frequency is increased and Memory Model is kept constant(SINGLE_CHANNEL_DDR3_1600).

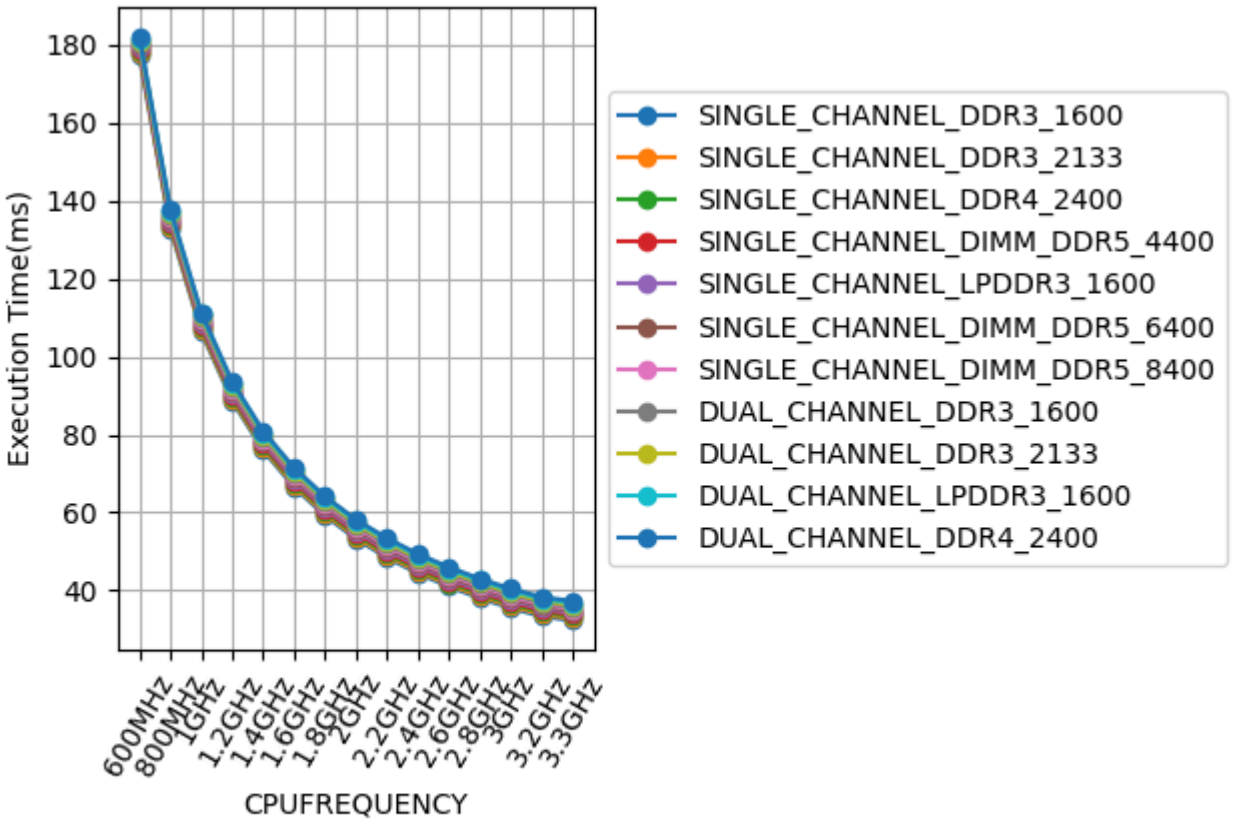


Figure 7: CPUFREQUENCY vs SIMSECONDS for different MEMORYMODEL(TIMING). Added jitter(0.5) to show overlapping line graphs.

- In Figure 7, Execution time(SimSeconds) decreases as CPU frequency is increased for different Memory models.

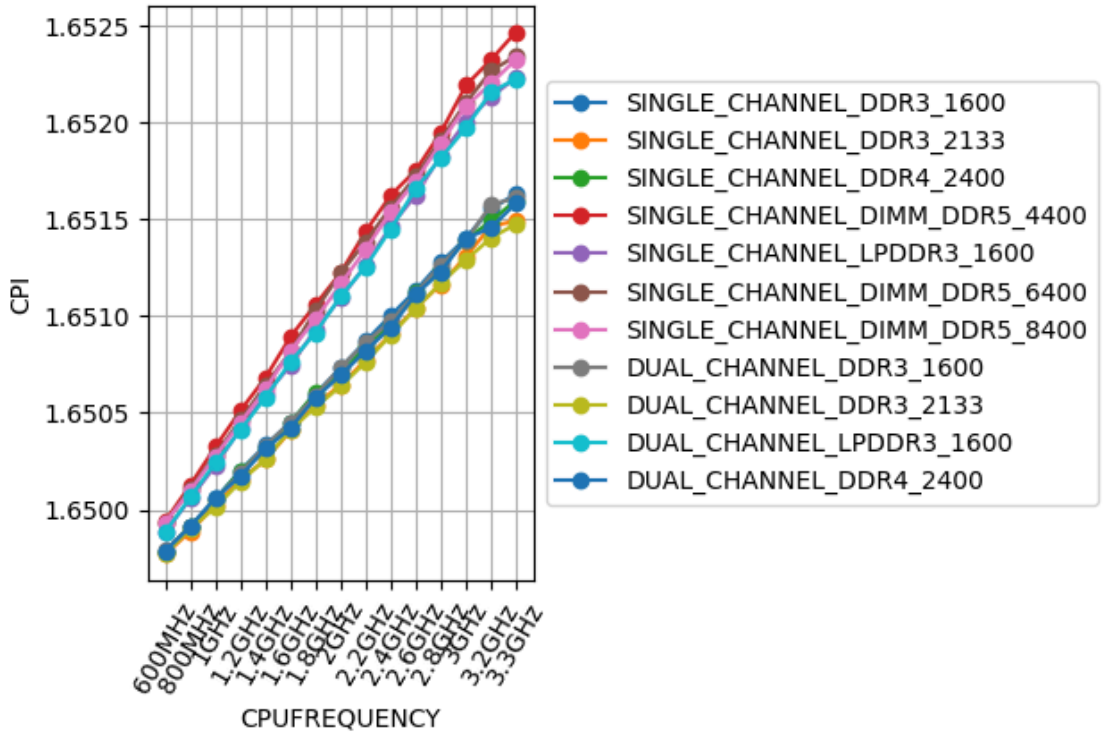


Figure 8: CPUFREQUENCY vs CPI for different MEMORYMODEL(TIMING).

- In Figure 8, the CPI for fixed TIMING CPU and different Memory Models stay close to 1.65 as CPU frequency is increased. No dramatic change in CPI.

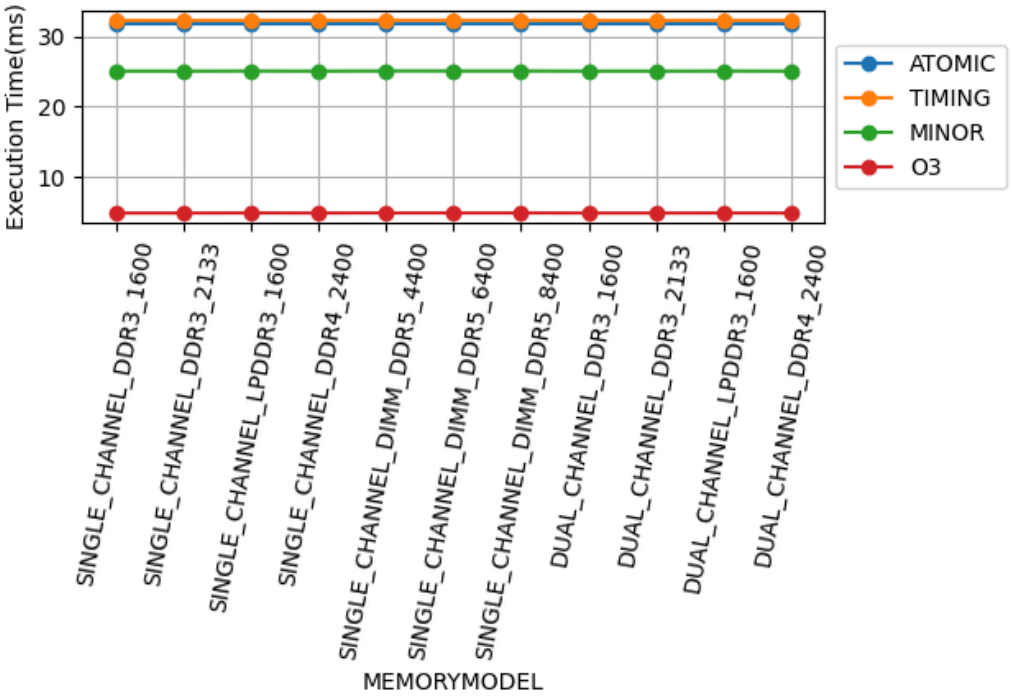


Figure 9: MEMORYMODEL vs SIMSECONDS for different CPUTYPE(3.3GHz).

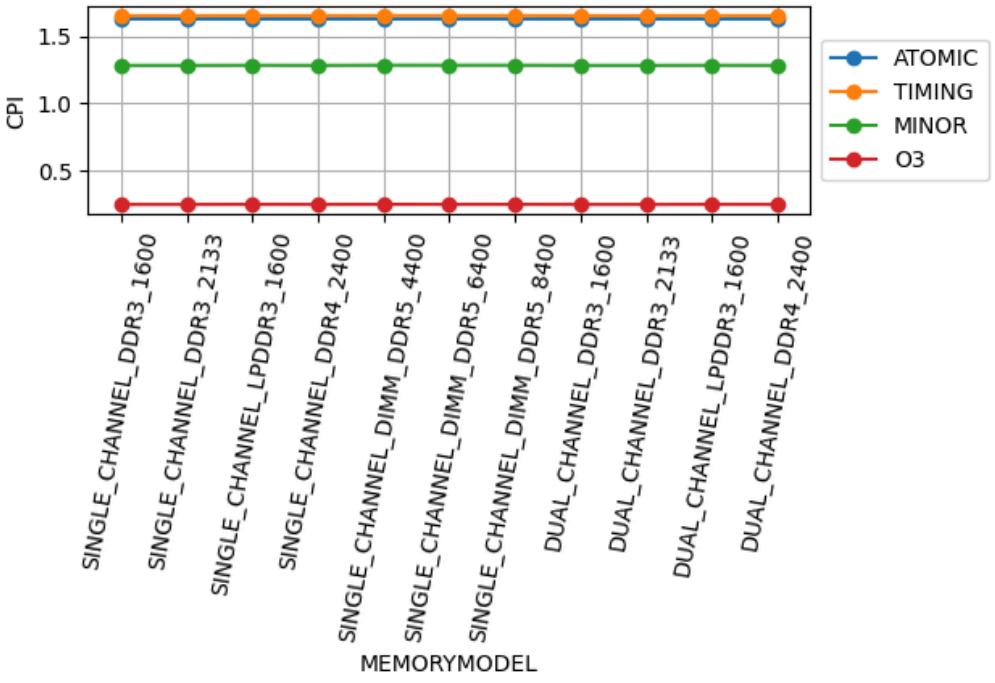


Figure 10: MEMORYMODEL vs CPI for different CPUTYPE(3.3GHz).

- According to Figures 9 and 10, changing Memory Models for all four types of CPUs has no impact on Execution time(simSeconds) or CPI, while keeping the CPU frequency(3.3 GHz) constant.

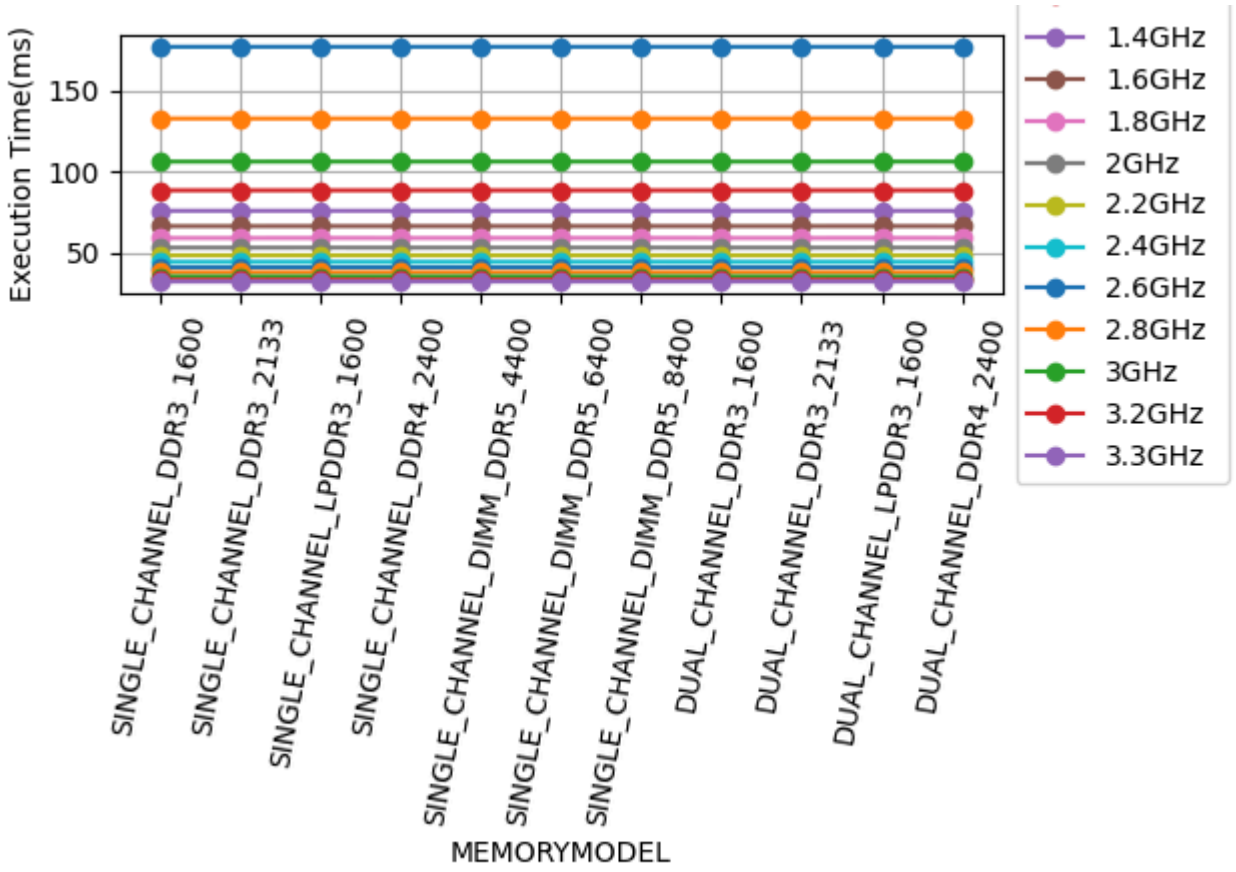


Figure 11: MEMORYMODEL vs SIMSECONDS for different CPUFREQUENCY(TIMING).

- According to Figure 11, changing Memory Models while keeping TIMING CPU constant, has no impact on Execution time. Higher CPU Frequencies leads to less execution time.

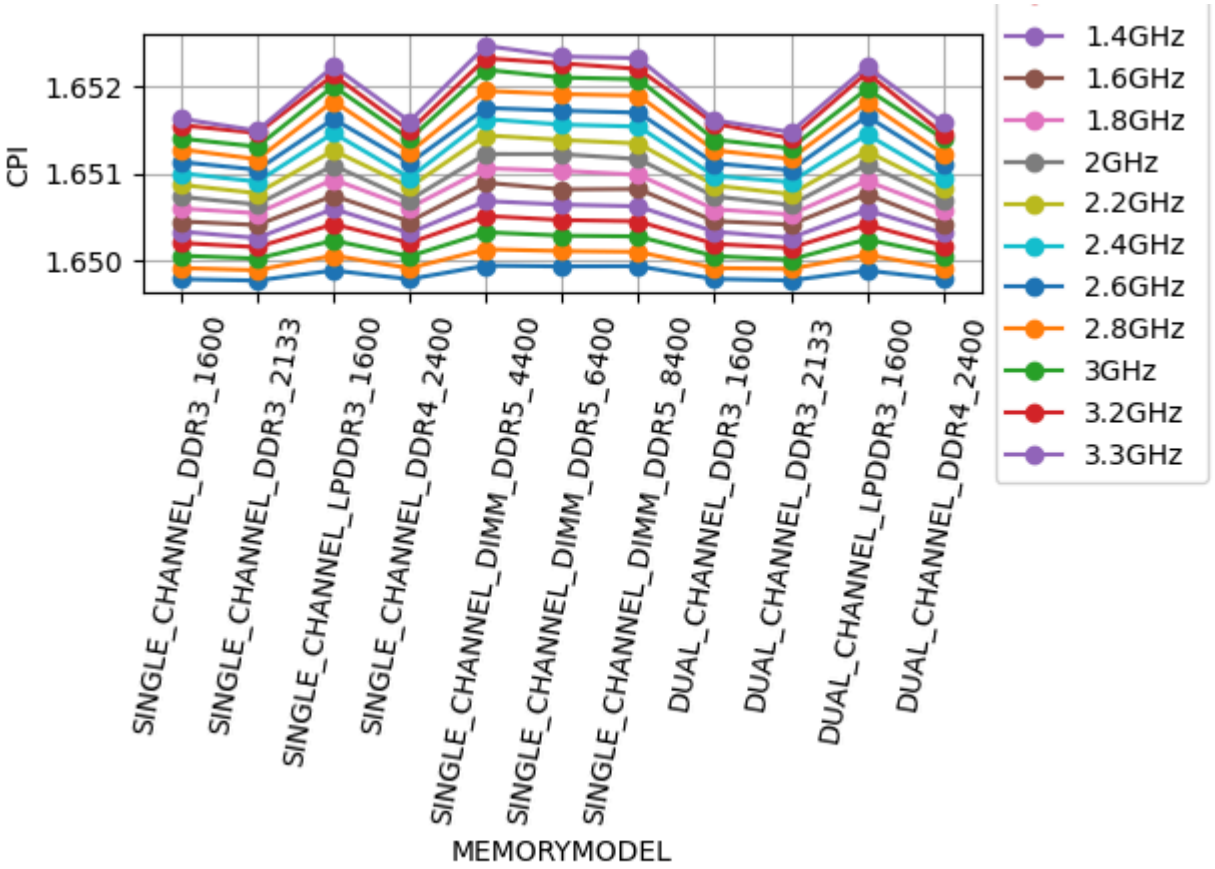


Figure 12: MEMORYMODEL vs CPI for different CPUFREQUENCY(TIMING).

- For Figure 12, the CPI for all memory models stays around ~1.65, for some Memory Models CPI increases(LPDDR3 RAMs) slightly for constant frequency and CPU type. This happens because of high latency for these specific models in comparison.

Conclusion

The following conclusions are drawn from observing simulation results of running cache optimized (L1d cache) Matrix Multiplication on various CPU types, CPU frequencies and Memory Models

- When comparing CPU Types, the four tested options can be stated below in increasing performance(less CPI and less execution time)

  TIMING CPU << ATOMIC CPU << MINOR CPU << O3 CPU

  - The above order is based on the following: Timing CPU performs in-order processing with simulation of memory access, Atomic CPU does not simulate proper memory access with in-order processing, Minor CPU has a 4-stage in-order pipeline, and finally, O3 CPU performs superscalar out-of-order processing with a 5-stage pipeline.

- When comparing CPU frequencies, increasing frequency for Atomic, Timing, and Minor brings down Execution time a lot more compared to O3 CPU, where the drop in execution time is not as significant(Figure 5)
- Varying Memory Models for a CPU type or CPU frequency does not have a significant impact on CPI or execution time, as the memory being used is not extremely large.

References

1) https://www.gem5.org/

ANNEXURE

A.1 Matrix Multiplication Code (L1d cache optimized)

```c
#include<stdio.h>
#include<stdlib.h>

#define MATRIXSIZE 100

unsigned int matrix1[MATRIXSIZE][MATRIXSIZE], matrix2[MATRIXSIZE][MATRIXSIZE];
unsigned long int resultmatrix[MATRIXSIZE][MATRIXSIZE];

int main(){


    //printf("Matrix Size is %d ", MATRIXSIZE);
    for(int i=0; i<MATRIXSIZE;++i){

        for(int j=0; j<MATRIXSIZE;++j){

            matrix1[i][j] = rand();
            matrix2[i][j] = rand();
            resultmatrix[i][j] = 0;
            //printf("(%d,%d) : %d %d\n",i,j, matrix1[i][j], matrix2[i][j]);
        }
    }

    // Transposing Second Matrix
    for(int i=0;i<MATRIXSIZE;++i){

        for(int j = 0; j < MATRIXSIZE; ++j){

        unsigned int temp = matrix2[i][j];
            matrix2[i][j] = matrix2[j][i];
            matrix2[j][i] = temp;
        }
    }

    for (int i = 0; i < MATRIXSIZE; ++i){
        for(int j = 0; j < MATRIXSIZE; ++j){

        unsigned long int sum = 0;
            for (int k = 0; k <MATRIXSIZE; ++k){
                resultmatrix[i][j] += matrix1[i][k] * matrix2[j][k];
        }

        }
    }


    return 0;
}
```

A.2 Benchmarking data used to construct all Graphs in the report.

Sno	ISA	CPUTYPE	CPUFREQUENCY	MEMORYMODEL	SIMSECONDS(s)	CPI
1	X86	ATOMIC	600MHz	SINGLE_CHANNEL_DDR3_2133	0.174862	1.628766
2	X86	ATOMIC	600MHz	SINGLE_CHANNEL_DDR4_2400	0.174862	1.628766
3	X86	ATOMIC	600MHz	SINGLE_CHANNEL_LPDDR3_1600	0.174862	1.628766
4	X86	ATOMIC	600MHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.174862	1.628766
5	X86	ATOMIC	600MHz	DUAL_CHANNEL_DDR3_1600	0.174862	1.628766
6	X86	ATOMIC	600MHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.174862	1.628766
7	X86	ATOMIC	600MHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.174862	1.628766
8	X86	ATOMIC	600MHz	DUAL_CHANNEL_DDR3_2133	0.174862	1.628766
9	X86	ATOMIC	600MHz	SINGLE_CHANNEL_DDR3_1600	0.174862	1.628766
10	X86	ATOMIC	600MHz	DUAL_CHANNEL_DDR4_2400	0.174862	1.628766
11	X86	ATOMIC	600MHz	DUAL_CHANNEL_LPDDR3_1600	0.174862	1.628766
12	X86	ATOMIC	800MHz	SINGLE_CHANNEL_DDR3_1600	0.131121	1.628766
13	X86	ATOMIC	800MHz	SINGLE_CHANNEL_DDR3_2133	0.131121	1.628766
14	X86	ATOMIC	800MHz	SINGLE_CHANNEL_LPDDR3_1600	0.131121	1.628766
15	X86	ATOMIC	800MHz	SINGLE_CHANNEL_DDR4_2400	0.131121	1.628766
16	X86	ATOMIC	800MHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.131121	1.628766
17	X86	ATOMIC	800MHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.131121	1.628766
18	X86	ATOMIC	800MHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.131121	1.628766
19	X86	ATOMIC	800MHz	DUAL_CHANNEL_DDR3_1600	0.131121	1.628766
20	X86	ATOMIC	800MHz	DUAL_CHANNEL_DDR3_2133	0.131121	1.628766
21	X86	ATOMIC	800MHz	DUAL_CHANNEL_LPDDR3_1600	0.131121	1.628766
22	X86	ATOMIC	800MHz	DUAL_CHANNEL_DDR4_2400	0.131121	1.628766
23	X86	ATOMIC	1GHz	SINGLE_CHANNEL_DDR3_1600	0.104896	1.628766
24	X86	ATOMIC	1GHz	SINGLE_CHANNEL_LPDDR3_1600	0.104896	1.628766
25	X86	ATOMIC	1GHz	SINGLE_CHANNEL_DDR3_2133	0.104896	1.628766
26	X86	ATOMIC	1GHz	SINGLE_CHANNEL_DDR4_2400	0.104896	1.628766
27	X86	ATOMIC	1GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.104896	1.628766
28	X86	ATOMIC	1GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.104896	1.628766
29	X86	ATOMIC	1GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.104896	1.628766
30	X86	ATOMIC	1GHz	DUAL_CHANNEL_DDR3_1600	0.104896	1.628766
31	X86	ATOMIC	1GHz	DUAL_CHANNEL_DDR3_2133	0.104896	1.628766
32	X86	ATOMIC	1GHz	DUAL_CHANNEL_LPDDR3_1600	0.104896	1.628766
33	X86	ATOMIC	1GHz	DUAL_CHANNEL_DDR4_2400	0.104896	1.628766
34	X86	ATOMIC	1.2GHz	SINGLE_CHANNEL_DDR3_1600	0.087379	1.628766
35	X86	ATOMIC	1.2GHz	SINGLE_CHANNEL_DDR3_2133	0.087379	1.628766
36	X86	ATOMIC	1.2GHz	SINGLE_CHANNEL_DDR4_2400	0.087379	1.628766
37	X86	ATOMIC	1.2GHz	SINGLE_CHANNEL_LPDDR3_1600	0.087379	1.628766
38	X86	ATOMIC	1.2GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.087379	1.628766
39	X86	ATOMIC	1.2GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.087379	1.628766

40	X86	ATOMIC	1.2GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.087379	1.628766
41	X86	ATOMIC	1.2GHz	DUAL_CHANNEL_DDR3_1600	0.087379	1.628766
42	X86	ATOMIC	1.2GHz	DUAL_CHANNEL_DDR3_2133	0.087379	1.628766
43	X86	ATOMIC	1.2GHz	DUAL_CHANNEL_LPDDR3_1600	0.087379	1.628766
44	X86	ATOMIC	1.2GHz	DUAL_CHANNEL_DDR4_2400	0.087379	1.628766
45	X86	ATOMIC	1.4GHz	SINGLE_CHANNEL_DDR3_1600	0.074896	1.628766
46	X86	ATOMIC	1.4GHz	SINGLE_CHANNEL_DDR3_2133	0.074896	1.628766
47	X86	ATOMIC	1.4GHz	SINGLE_CHANNEL_LPDDR3_1600	0.074896	1.628766
48	X86	ATOMIC	1.4GHz	SINGLE_CHANNEL_DDR4_2400	0.074896	1.628766
49	X86	ATOMIC	1.4GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.074896	1.628766
50	X86	ATOMIC	1.4GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.074896	1.628766
51	X86	ATOMIC	1.4GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.074896	1.628766
52	X86	ATOMIC	1.4GHz	DUAL_CHANNEL_DDR3_1600	0.074896	1.628766
53	X86	ATOMIC	1.4GHz	DUAL_CHANNEL_DDR3_2133	0.074896	1.628766
54	X86	ATOMIC	1.4GHz	DUAL_CHANNEL_LPDDR3_1600	0.074896	1.628766
55	X86	ATOMIC	1.4GHz	DUAL_CHANNEL_DDR4_2400	0.074896	1.628766
56	X86	ATOMIC	1.6GHz	SINGLE_CHANNEL_DDR3_2133	0.06556	1.628766
57	X86	ATOMIC	1.6GHz	SINGLE_CHANNEL_LPDDR3_1600	0.06556	1.628766
58	X86	ATOMIC	1.6GHz	SINGLE_CHANNEL_DDR3_1600	0.06556	1.628766
59	X86	ATOMIC	1.6GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.06556	1.628766
60	X86	ATOMIC	1.6GHz	SINGLE_CHANNEL_DDR4_2400	0.06556	1.628766
61	X86	ATOMIC	1.6GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.06556	1.628766
62	X86	ATOMIC	1.6GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.06556	1.628766
63	X86	ATOMIC	1.6GHz	DUAL_CHANNEL_DDR3_1600	0.06556	1.628766
64	X86	ATOMIC	1.6GHz	DUAL_CHANNEL_DDR3_2133	0.06556	1.628766
65	X86	ATOMIC	1.6GHz	DUAL_CHANNEL_LPDDR3_1600	0.06556	1.628766
66	X86	ATOMIC	1.6GHz	DUAL_CHANNEL_DDR4_2400	0.06556	1.628766
67	X86	ATOMIC	1.8GHz	SINGLE_CHANNEL_LPDDR3_1600	0.058322	1.628766
68	X86	ATOMIC	1.8GHz	SINGLE_CHANNEL_DDR3_2133	0.058322	1.628766
69	X86	ATOMIC	1.8GHz	SINGLE_CHANNEL_DDR4_2400	0.058322	1.628766
70	X86	ATOMIC	1.8GHz	SINGLE_CHANNEL_DDR3_1600	0.058322	1.628766
71	X86	ATOMIC	1.8GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.058322	1.628766
72	X86	ATOMIC	1.8GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.058322	1.628766
73	X86	ATOMIC	1.8GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.058322	1.628766
74	X86	ATOMIC	1.8GHz	DUAL_CHANNEL_DDR3_1600	0.058322	1.628766
75	X86	ATOMIC	1.8GHz	DUAL_CHANNEL_DDR3_2133	0.058322	1.628766
76	X86	ATOMIC	1.8GHz	DUAL_CHANNEL_LPDDR3_1600	0.058322	1.628766
77	X86	ATOMIC	1.8GHz	DUAL_CHANNEL_DDR4_2400	0.058322	1.628766
78	X86	ATOMIC	2GHz	SINGLE_CHANNEL_DDR3_2133	0.052448	1.628766
79	X86	ATOMIC	2GHz	SINGLE_CHANNEL_DDR3_1600	0.052448	1.628766
80	X86	ATOMIC	2GHz	SINGLE_CHANNEL_DDR4_2400	0.052448	1.628766
81	X86	ATOMIC	2GHz	SINGLE_CHANNEL_LPDDR3_1600	0.052448	1.628766
82	X86	ATOMIC	2GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.052448	1.628766
83	X86	ATOMIC	2GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.052448	1.628766

84	X86	ATOMIC	2GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.052448	1.628766
85	X86	ATOMIC	2GHz	DUAL_CHANNEL_DDR3_1600	0.052448	1.628766
86	X86	ATOMIC	2GHz	DUAL_CHANNEL_DDR3_2133	0.052448	1.628766
87	X86	ATOMIC	2GHz	DUAL_CHANNEL_LPDDR3_1600	0.052448	1.628766
88	X86	ATOMIC	2GHz	DUAL_CHANNEL_DDR4_2400	0.052448	1.628766
89	X86	ATOMIC	2.2GHz	SINGLE_CHANNEL_DDR3_1600	0.047728	1.628766
90	X86	ATOMIC	2.2GHz	SINGLE_CHANNEL_LPDDR3_1600	0.047728	1.628766
91	X86	ATOMIC	2.2GHz	SINGLE_CHANNEL_DDR3_2133	0.047728	1.628766
92	X86	ATOMIC	2.2GHz	SINGLE_CHANNEL_DDR4_2400	0.047728	1.628766
93	X86	ATOMIC	2.2GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.047728	1.628766
94	X86	ATOMIC	2.2GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.047728	1.628766
95	X86	ATOMIC	2.2GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.047728	1.628766
96	X86	ATOMIC	2.2GHz	DUAL_CHANNEL_DDR3_1600	0.047728	1.628766
97	X86	ATOMIC	2.2GHz	DUAL_CHANNEL_DDR3_2133	0.047728	1.628766
98	X86	ATOMIC	2.2GHz	DUAL_CHANNEL_LPDDR3_1600	0.047728	1.628766
99	X86	ATOMIC	2.2GHz	DUAL_CHANNEL_DDR4_2400	0.047728	1.628766
100	X86	ATOMIC	2.4GHz	SINGLE_CHANNEL_DDR4_2400	0.043742	1.628766
101	X86	ATOMIC	2.4GHz	SINGLE_CHANNEL_DDR3_2133	0.043742	1.628766
102	X86	ATOMIC	2.4GHz	SINGLE_CHANNEL_LPDDR3_1600	0.043742	1.628766
103	X86	ATOMIC	2.4GHz	SINGLE_CHANNEL_DDR3_1600	0.043742	1.628766
104	X86	ATOMIC	2.4GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.043742	1.628766
105	X86	ATOMIC	2.4GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.043742	1.628766
106	X86	ATOMIC	2.4GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.043742	1.628766
107	X86	ATOMIC	2.4GHz	DUAL_CHANNEL_DDR3_1600	0.043742	1.628766
108	X86	ATOMIC	2.4GHz	DUAL_CHANNEL_DDR3_2133	0.043742	1.628766
109	X86	ATOMIC	2.4GHz	DUAL_CHANNEL_LPDDR3_1600	0.043742	1.628766
110	X86	ATOMIC	2.4GHz	DUAL_CHANNEL_DDR4_2400	0.043742	1.628766
111	X86	ATOMIC	2.6GHz	SINGLE_CHANNEL_LPDDR3_1600	0.040385	1.628766
112	X86	ATOMIC	2.6GHz	SINGLE_CHANNEL_DDR3_1600	0.040385	1.628766
113	X86	ATOMIC	2.6GHz	SINGLE_CHANNEL_DDR3_2133	0.040385	1.628766
114	X86	ATOMIC	2.6GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.040385	1.628766
115	X86	ATOMIC	2.6GHz	SINGLE_CHANNEL_DDR4_2400	0.040385	1.628766
116	X86	ATOMIC	2.6GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.040385	1.628766
117	X86	ATOMIC	2.6GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.040385	1.628766
118	X86	ATOMIC	2.6GHz	DUAL_CHANNEL_DDR3_1600	0.040385	1.628766
119	X86	ATOMIC	2.6GHz	DUAL_CHANNEL_DDR3_2133	0.040385	1.628766
120	X86	ATOMIC	2.6GHz	DUAL_CHANNEL_LPDDR3_1600	0.040385	1.628766
121	X86	ATOMIC	2.6GHz	DUAL_CHANNEL_DDR4_2400	0.040385	1.628766
122	X86	ATOMIC	2.8GHz	SINGLE_CHANNEL_DDR3_1600	0.037448	1.628766
123	X86	ATOMIC	2.8GHz	SINGLE_CHANNEL_LPDDR3_1600	0.037448	1.628766
124	X86	ATOMIC	2.8GHz	SINGLE_CHANNEL_DDR4_2400	0.037448	1.628766
125	X86	ATOMIC	2.8GHz	SINGLE_CHANNEL_DDR3_2133	0.037448	1.628766
126	X86	ATOMIC	2.8GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.037448	1.628766
127	X86	ATOMIC	2.8GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.037448	1.628766

128	X86	ATOMIC	2.8GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.037448	1.628766
129	X86	ATOMIC	2.8GHz	DUAL_CHANNEL_DDR3_1600	0.037448	1.628766
130	X86	ATOMIC	2.8GHz	DUAL_CHANNEL_DDR3_2133	0.037448	1.628766
131	X86	ATOMIC	2.8GHz	DUAL_CHANNEL_LPDDR3_1600	0.037448	1.628766
132	X86	ATOMIC	2.8GHz	DUAL_CHANNEL_DDR4_2400	0.037448	1.628766
133	X86	ATOMIC	3GHz	SINGLE_CHANNEL_DDR3_1600	0.034931	1.628766
134	X86	ATOMIC	3GHz	SINGLE_CHANNEL_DDR3_2133	0.034931	1.628766
135	X86	ATOMIC	3GHz	SINGLE_CHANNEL_LPDDR3_1600	0.034931	1.628766
136	X86	ATOMIC	3GHz	SINGLE_CHANNEL_DDR4_2400	0.034931	1.628766
137	X86	ATOMIC	3GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.034931	1.628766
138	X86	ATOMIC	3GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.034931	1.628766
139	X86	ATOMIC	3GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.034931	1.628766
140	X86	ATOMIC	3GHz	DUAL_CHANNEL_DDR3_1600	0.034931	1.628766
141	X86	ATOMIC	3GHz	DUAL_CHANNEL_DDR3_2133	0.034931	1.628766
142	X86	ATOMIC	3GHz	DUAL_CHANNEL_LPDDR3_1600	0.034931	1.628766
143	X86	ATOMIC	3GHz	DUAL_CHANNEL_DDR4_2400	0.034931	1.628766
144	X86	ATOMIC	3.2GHz	SINGLE_CHANNEL_DDR3_2133	0.032833	1.628766
145	X86	ATOMIC	3.2GHz	SINGLE_CHANNEL_DDR3_1600	0.032833	1.628766
146	X86	ATOMIC	3.2GHz	SINGLE_CHANNEL_LPDDR3_1600	0.032833	1.628766
147	X86	ATOMIC	3.2GHz	SINGLE_CHANNEL_DDR4_2400	0.032833	1.628766
148	X86	ATOMIC	3.2GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.032833	1.628766
149	X86	ATOMIC	3.2GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.032833	1.628766
150	X86	ATOMIC	3.2GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.032833	1.628766
151	X86	ATOMIC	3.2GHz	DUAL_CHANNEL_DDR3_1600	0.032833	1.628766
152	X86	ATOMIC	3.2GHz	DUAL_CHANNEL_DDR3_2133	0.032833	1.628766
153	X86	ATOMIC	3.2GHz	DUAL_CHANNEL_LPDDR3_1600	0.032833	1.628766
154	X86	ATOMIC	3.2GHz	DUAL_CHANNEL_DDR4_2400	0.032833	1.628766
155	X86	ATOMIC	3.3GHz	SINGLE_CHANNEL_DDR3_2133	0.031784	1.628766
156	X86	ATOMIC	3.3GHz	SINGLE_CHANNEL_LPDDR3_1600	0.031784	1.628766
157	X86	ATOMIC	3.3GHz	SINGLE_CHANNEL_DDR3_1600	0.031784	1.628766
158	X86	ATOMIC	3.3GHz	SINGLE_CHANNEL_DDR4_2400	0.031784	1.628766
159	X86	ATOMIC	3.3GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.031784	1.628766
160	X86	ATOMIC	3.3GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.031784	1.628766
161	X86	ATOMIC	3.3GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.031784	1.628766
162	X86	ATOMIC	3.3GHz	DUAL_CHANNEL_DDR3_1600	0.031784	1.628766
163	X86	ATOMIC	3.3GHz	DUAL_CHANNEL_DDR3_2133	0.031784	1.628766
164	X86	ATOMIC	3.3GHz	DUAL_CHANNEL_LPDDR3_1600	0.031784	1.628766
165	X86	ATOMIC	3.3GHz	DUAL_CHANNEL_DDR4_2400	0.031784	1.628766
166	X86	TIMING	600MHz	SINGLE_CHANNEL_DDR3_1600	0.177254	1.649787
167	X86	TIMING	600MHz	SINGLE_CHANNEL_DDR3_2133	0.177253	1.64977
168	X86	TIMING	600MHz	SINGLE_CHANNEL_DDR4_2400	0.177254	1.649786
169	X86	TIMING	600MHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.177271	1.649942
170	X86	TIMING	600MHz	SINGLE_CHANNEL_LPDDR3_1600	0.177265	1.649886
171	X86	TIMING	600MHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.17727	1.649933

172	X86	TIMING	600MHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.17727	1.649935
173	X86	TIMING	600MHz	DUAL_CHANNEL_DDR3_1600	0.177255	1.649789
174	X86	TIMING	600MHz	DUAL_CHANNEL_DDR3_2133	0.177253	1.649772
175	X86	TIMING	600MHz	DUAL_CHANNEL_LPDDR3_1600	0.177265	1.649886
176	X86	TIMING	600MHz	DUAL_CHANNEL_DDR4_2400	0.177254	1.649786
177	X86	TIMING	800MHz	SINGLE_CHANNEL_DDR3_1600	0.132924	1.649914
178	X86	TIMING	800MHz	SINGLE_CHANNEL_DDR3_2133	0.132922	1.649889
179	X86	TIMING	800MHz	SINGLE_CHANNEL_LPDDR3_1600	0.132936	1.650058
180	X86	TIMING	800MHz	SINGLE_CHANNEL_DDR4_2400	0.132924	1.649913
181	X86	TIMING	800MHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.132942	1.650129
182	X86	TIMING	800MHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.13294	1.650109
183	X86	TIMING	800MHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.132939	1.6501
184	X86	TIMING	800MHz	DUAL_CHANNEL_DDR3_1600	0.132924	1.649913
185	X86	TIMING	800MHz	DUAL_CHANNEL_DDR3_2133	0.132924	1.649904
186	X86	TIMING	800MHz	DUAL_CHANNEL_LPDDR3_1600	0.132937	1.650065
187	X86	TIMING	800MHz	DUAL_CHANNEL_DDR4_2400	0.132924	1.649914
188	X86	TIMING	1GHz	SINGLE_CHANNEL_DDR3_1600	0.106349	1.650056
189	X86	TIMING	1GHz	SINGLE_CHANNEL_DDR3_2133	0.106347	1.650023
190	X86	TIMING	1GHz	SINGLE_CHANNEL_LPDDR3_1600	0.10636	1.650229
191	X86	TIMING	1GHz	SINGLE_CHANNEL_DDR4_2400	0.106348	1.650049
192	X86	TIMING	1GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.106366	1.650325
193	X86	TIMING	1GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.106364	1.650286
194	X86	TIMING	1GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.106363	1.650277
195	X86	TIMING	1GHz	DUAL_CHANNEL_DDR3_1600	0.106348	1.650051
196	X86	TIMING	1GHz	DUAL_CHANNEL_DDR3_2133	0.106346	1.650012
197	X86	TIMING	1GHz	DUAL_CHANNEL_LPDDR3_1600	0.106361	1.650243
198	X86	TIMING	1GHz	DUAL_CHANNEL_DDR4_2400	0.106349	1.65006
199	X86	TIMING	1.2GHz	SINGLE_CHANNEL_DDR3_1600	0.088596	1.650202
200	X86	TIMING	1.2GHz	SINGLE_CHANNEL_DDR3_2133	0.088594	1.650154
201	X86	TIMING	1.2GHz	SINGLE_CHANNEL_LPDDR3_1600	0.088608	1.650418
202	X86	TIMING	1.2GHz	SINGLE_CHANNEL_DDR4_2400	0.088596	1.650202
203	X86	TIMING	1.2GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.088613	1.650512
204	X86	TIMING	1.2GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.08861	1.650465
205	X86	TIMING	1.2GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.08861	1.650451
206	X86	TIMING	1.2GHz	DUAL_CHANNEL_DDR3_1600	0.088596	1.650191
207	X86	TIMING	1.2GHz	DUAL_CHANNEL_DDR3_2133	0.088593	1.650146
208	X86	TIMING	1.2GHz	DUAL_CHANNEL_LPDDR3_1600	0.088608	1.650412
209	X86	TIMING	1.2GHz	DUAL_CHANNEL_DDR4_2400	0.088595	1.650173
210	X86	TIMING	1.4GHz	SINGLE_CHANNEL_DDR3_1600	0.075946	1.650333
211	X86	TIMING	1.4GHz	SINGLE_CHANNEL_LPDDR3_1600	0.075958	1.650594
212	X86	TIMING	1.4GHz	SINGLE_CHANNEL_DDR3_2133	0.075942	1.650259
213	X86	TIMING	1.4GHz	SINGLE_CHANNEL_DDR4_2400	0.075945	1.650318
214	X86	TIMING	1.4GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.075962	1.650683
215	X86	TIMING	1.4GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.07596	1.650643

216	X86	TIMING	1.4GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.075959	1.650623
217	X86	TIMING	1.4GHz	DUAL_CHANNEL_DDR3_1600	0.075946	1.650331
218	X86	TIMING	1.4GHz	DUAL_CHANNEL_DDR3_2133	0.075942	1.650261
219	X86	TIMING	1.4GHz	DUAL_CHANNEL_LPDDR3_1600	0.075957	1.65058
220	X86	TIMING	1.4GHz	DUAL_CHANNEL_DDR4_2400	0.075945	1.650315
221	X86	TIMING	1.6GHz	SINGLE_CHANNEL_DDR3_1600	0.066484	1.65045
222	X86	TIMING	1.6GHz	SINGLE_CHANNEL_DDR3_2133	0.066482	1.650408
223	X86	TIMING	1.6GHz	SINGLE_CHANNEL_DDR4_2400	0.066484	1.650449
224	X86	TIMING	1.6GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.066502	1.650894
225	X86	TIMING	1.6GHz	SINGLE_CHANNEL_LPDDR3_1600	0.066496	1.650746
226	X86	TIMING	1.6GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.066499	1.650817
227	X86	TIMING	1.6GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.066499	1.650822
228	X86	TIMING	1.6GHz	DUAL_CHANNEL_DDR3_1600	0.066484	1.650453
229	X86	TIMING	1.6GHz	DUAL_CHANNEL_DDR3_2133	0.066482	1.650414
230	X86	TIMING	1.6GHz	DUAL_CHANNEL_LPDDR3_1600	0.066496	1.650764
231	X86	TIMING	1.6GHz	DUAL_CHANNEL_DDR4_2400	0.066483	1.650421
232	X86	TIMING	1.8GHz	SINGLE_CHANNEL_DDR3_1600	0.059149	1.650598
233	X86	TIMING	1.8GHz	SINGLE_CHANNEL_DDR3_2133	0.059147	1.650543
234	X86	TIMING	1.8GHz	SINGLE_CHANNEL_LPDDR3_1600	0.059161	1.650932
235	X86	TIMING	1.8GHz	SINGLE_CHANNEL_DDR4_2400	0.05915	1.650606
236	X86	TIMING	1.8GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.059166	1.65106
237	X86	TIMING	1.8GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.059165	1.651031
238	X86	TIMING	1.8GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.059163	1.650985
239	X86	TIMING	1.8GHz	DUAL_CHANNEL_DDR3_1600	0.059149	1.650589
240	X86	TIMING	1.8GHz	DUAL_CHANNEL_DDR3_2133	0.059147	1.65053
241	X86	TIMING	1.8GHz	DUAL_CHANNEL_LPDDR3_1600	0.059161	1.650914
242	X86	TIMING	1.8GHz	DUAL_CHANNEL_DDR4_2400	0.059149	1.650579
243	X86	TIMING	2GHz	SINGLE_CHANNEL_DDR3_2133	0.053193	1.650644
244	X86	TIMING	2GHz	SINGLE_CHANNEL_DDR3_1600	0.053196	1.650732
245	X86	TIMING	2GHz	SINGLE_CHANNEL_LPDDR3_1600	0.053208	1.651092
246	X86	TIMING	2GHz	SINGLE_CHANNEL_DDR4_2400	0.053195	1.650698
247	X86	TIMING	2GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.053212	1.651223
248	X86	TIMING	2GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.053212	1.651222
249	X86	TIMING	2GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.05321	1.651165
250	X86	TIMING	2GHz	DUAL_CHANNEL_DDR3_1600	0.053196	1.650737
251	X86	TIMING	2GHz	DUAL_CHANNEL_DDR3_2133	0.053193	1.650638
252	X86	TIMING	2GHz	DUAL_CHANNEL_LPDDR3_1600	0.053208	1.651103
253	X86	TIMING	2GHz	DUAL_CHANNEL_DDR4_2400	0.053195	1.650694
254	X86	TIMING	2.2GHz	SINGLE_CHANNEL_DDR3_1600	0.048413	1.65087
255	X86	TIMING	2.2GHz	SINGLE_CHANNEL_LPDDR3_1600	0.048424	1.651264
256	X86	TIMING	2.2GHz	SINGLE_CHANNEL_DDR3_2133	0.04841	1.650773
257	X86	TIMING	2.2GHz	SINGLE_CHANNEL_DDR4_2400	0.048412	1.650845
258	X86	TIMING	2.2GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.048429	1.651438
259	X86	TIMING	2.2GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.048428	1.651385

260	X86	TIMING	2.2GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.048426	1.651345
261	X86	TIMING	2.2GHz	DUAL_CHANNEL_DDR3_1600	0.048412	1.650863
262	X86	TIMING	2.2GHz	DUAL_CHANNEL_DDR3_2133	0.048409	1.650762
263	X86	TIMING	2.2GHz	DUAL_CHANNEL_LPDDR3_1600	0.048424	1.651255
264	X86	TIMING	2.2GHz	DUAL_CHANNEL_DDR4_2400	0.048411	1.650821
265	X86	TIMING	2.4GHz	SINGLE_CHANNEL_DDR3_2133	0.04437	1.650906
266	X86	TIMING	2.4GHz	SINGLE_CHANNEL_DDR3_1600	0.044373	1.651003
267	X86	TIMING	2.4GHz	SINGLE_CHANNEL_LPDDR3_1600	0.044385	1.651464
268	X86	TIMING	2.4GHz	SINGLE_CHANNEL_DDR4_2400	0.044372	1.650953
269	X86	TIMING	2.4GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.044388	1.651562
270	X86	TIMING	2.4GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.04439	1.651624
271	X86	TIMING	2.4GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.044387	1.651539
272	X86	TIMING	2.4GHz	DUAL_CHANNEL_DDR3_1600	0.044372	1.650973
273	X86	TIMING	2.4GHz	DUAL_CHANNEL_DDR3_2133	0.04437	1.650902
274	X86	TIMING	2.4GHz	DUAL_CHANNEL_LPDDR3_1600	0.044385	1.651449
275	X86	TIMING	2.4GHz	DUAL_CHANNEL_DDR4_2400	0.044371	1.650941
276	X86	TIMING	2.6GHz	SINGLE_CHANNEL_DDR3_1600	0.040971	1.651132
277	X86	TIMING	2.6GHz	SINGLE_CHANNEL_DDR3_2133	0.040969	1.651043
278	X86	TIMING	2.6GHz	SINGLE_CHANNEL_LPDDR3_1600	0.040983	1.651625
279	X86	TIMING	2.6GHz	SINGLE_CHANNEL_DDR4_2400	0.040971	1.65112
280	X86	TIMING	2.6GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.040986	1.651722
281	X86	TIMING	2.6GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.040986	1.651753
282	X86	TIMING	2.6GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.040985	1.651699
283	X86	TIMING	2.6GHz	DUAL_CHANNEL_DDR3_1600	0.040971	1.651118
284	X86	TIMING	2.6GHz	DUAL_CHANNEL_DDR3_2133	0.040969	1.651039
285	X86	TIMING	2.6GHz	DUAL_CHANNEL_LPDDR3_1600	0.040984	1.651658
286	X86	TIMING	2.6GHz	DUAL_CHANNEL_DDR4_2400	0.04097	1.651112
287	X86	TIMING	2.8GHz	SINGLE_CHANNEL_DDR3_1600	0.037995	1.651277
288	X86	TIMING	2.8GHz	SINGLE_CHANNEL_LPDDR3_1600	0.038007	1.65183
289	X86	TIMING	2.8GHz	SINGLE_CHANNEL_DDR3_2133	0.037992	1.651164
290	X86	TIMING	2.8GHz	SINGLE_CHANNEL_DDR4_2400	0.037994	1.651245
291	X86	TIMING	2.8GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.03801	1.651948
292	X86	TIMING	2.8GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.038009	1.651912
293	X86	TIMING	2.8GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.038009	1.651894
294	X86	TIMING	2.8GHz	DUAL_CHANNEL_DDR3_1600	0.037994	1.651266
295	X86	TIMING	2.8GHz	DUAL_CHANNEL_DDR3_2133	0.037992	1.65117
296	X86	TIMING	2.8GHz	DUAL_CHANNEL_LPDDR3_1600	0.038007	1.651816
297	X86	TIMING	2.8GHz	DUAL_CHANNEL_DDR4_2400	0.037993	1.651229
298	X86	TIMING	3GHz	SINGLE_CHANNEL_DDR3_1600	0.035443	1.651402
299	X86	TIMING	3GHz	SINGLE_CHANNEL_DDR3_2133	0.035441	1.651308
300	X86	TIMING	3GHz	SINGLE_CHANNEL_LPDDR3_1600	0.035456	1.652002
301	X86	TIMING	3GHz	SINGLE_CHANNEL_DDR4_2400	0.035443	1.651395
302	X86	TIMING	3GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.03546	1.652191
303	X86	TIMING	3GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.035458	1.652102

304	X86	TIMING	3GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.035458	1.652083
305	X86	TIMING	3GHz	DUAL_CHANNEL_DDR3_1600	0.035443	1.651389
306	X86	TIMING	3GHz	DUAL_CHANNEL_DDR3_2133	0.035441	1.65129
307	X86	TIMING	3GHz	DUAL_CHANNEL_LPDDR3_1600	0.035455	1.651971
308	X86	TIMING	3GHz	DUAL_CHANNEL_DDR4_2400	0.035443	1.6514
309	X86	TIMING	3.2GHz	SINGLE_CHANNEL_DDR3_1600	0.033317	1.651555
310	X86	TIMING	3.2GHz	SINGLE_CHANNEL_DDR3_2133	0.033316	1.651464
311	X86	TIMING	3.2GHz	SINGLE_CHANNEL_DDR4_2400	0.033316	1.65149
312	X86	TIMING	3.2GHz	SINGLE_CHANNEL_LPDDR3_1600	0.033329	1.652131
313	X86	TIMING	3.2GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.033333	1.652323
314	X86	TIMING	3.2GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.033332	1.652265
315	X86	TIMING	3.2GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.03333	1.652201
316	X86	TIMING	3.2GHz	DUAL_CHANNEL_DDR3_1600	0.033318	1.651574
317	X86	TIMING	3.2GHz	DUAL_CHANNEL_DDR3_2133	0.033314	1.651403
318	X86	TIMING	3.2GHz	DUAL_CHANNEL_LPDDR3_1600	0.033329	1.652156
319	X86	TIMING	3.2GHz	DUAL_CHANNEL_DDR4_2400	0.033315	1.651456
320	X86	TIMING	3.3GHz	SINGLE_CHANNEL_DDR3_2133	0.032252	1.651491
321	X86	TIMING	3.3GHz	SINGLE_CHANNEL_LPDDR3_1600	0.032266	1.652232
322	X86	TIMING	3.3GHz	SINGLE_CHANNEL_DDR4_2400	0.032254	1.651598
323	X86	TIMING	3.3GHz	SINGLE_CHANNEL_DDR3_1600	0.032254	1.651629
324	X86	TIMING	3.3GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.032271	1.652467
325	X86	TIMING	3.3GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.032268	1.652344
326	X86	TIMING	3.3GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.032268	1.652325
327	X86	TIMING	3.3GHz	DUAL_CHANNEL_DDR3_1600	0.032254	1.651611
328	X86	TIMING	3.3GHz	DUAL_CHANNEL_DDR3_2133	0.032251	1.651475
329	X86	TIMING	3.3GHz	DUAL_CHANNEL_LPDDR3_1600	0.032266	1.652225
330	X86	TIMING	3.3GHz	DUAL_CHANNEL_DDR4_2400	0.032253	1.651582
331	X86	MINOR	600MHz	SINGLE_CHANNEL_DDR3_1600	0.137488	1.27966
332	X86	MINOR	600MHz	SINGLE_CHANNEL_DDR3_2133	0.137485	1.279631
333	X86	MINOR	600MHz	SINGLE_CHANNEL_LPDDR3_1600	0.137501	1.279784
334	X86	MINOR	600MHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.137511	1.279881
335	X86	MINOR	600MHz	SINGLE_CHANNEL_DDR4_2400	0.137487	1.279656
336	X86	MINOR	600MHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.13751	1.27987
337	X86	MINOR	600MHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.137511	1.279873
338	X86	MINOR	600MHz	DUAL_CHANNEL_DDR3_1600	0.137489	1.279668
339	X86	MINOR	600MHz	DUAL_CHANNEL_DDR3_2133	0.137485	1.27964
340	X86	MINOR	600MHz	DUAL_CHANNEL_LPDDR3_1600	0.1375	1.279779
341	X86	MINOR	600MHz	DUAL_CHANNEL_DDR4_2400	0.137489	1.279673
342	X86	MINOR	800MHz	SINGLE_CHANNEL_DDR3_1600	0.103108	1.279824
343	X86	MINOR	800MHz	SINGLE_CHANNEL_DDR3_2133	0.103106	1.279799
344	X86	MINOR	800MHz	SINGLE_CHANNEL_LPDDR3_1600	0.103125	1.280027
345	X86	MINOR	800MHz	SINGLE_CHANNEL_DDR4_2400	0.10311	1.27984
346	X86	MINOR	800MHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.103135	1.280161
347	X86	MINOR	800MHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.103133	1.280129

348	X86	MINOR	800MHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.103133	1.280135
349	X86	MINOR	800MHz	DUAL_CHANNEL_DDR3_1600	0.103109	1.279838
350	X86	MINOR	800MHz	DUAL_CHANNEL_DDR3_2133	0.103106	1.279797
351	X86	MINOR	800MHz	DUAL_CHANNEL_LPDDR3_1600	0.103124	1.280013
352	X86	MINOR	800MHz	DUAL_CHANNEL_DDR4_2400	0.10311	1.279841
353	X86	MINOR	1GHz	SINGLE_CHANNEL_LPDDR3_1600	0.082514	1.280254
354	X86	MINOR	1GHz	SINGLE_CHANNEL_DDR3_1600	0.0825	1.280024
355	X86	MINOR	1GHz	SINGLE_CHANNEL_DDR3_2133	0.082496	1.279972
356	X86	MINOR	1GHz	SINGLE_CHANNEL_DDR4_2400	0.0825	1.280027
357	X86	MINOR	1GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.082525	1.280418
358	X86	MINOR	1GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.082522	1.280365
359	X86	MINOR	1GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.082522	1.280365
360	X86	MINOR	1GHz	DUAL_CHANNEL_DDR3_1600	0.082499	1.280018
361	X86	MINOR	1GHz	DUAL_CHANNEL_DDR3_2133	0.082497	1.279985
362	X86	MINOR	1GHz	DUAL_CHANNEL_LPDDR3_1600	0.082514	1.280245
363	X86	MINOR	1GHz	DUAL_CHANNEL_DDR4_2400	0.082499	1.280021
364	X86	MINOR	1.2GHz	SINGLE_CHANNEL_LPDDR3_1600	0.06875	1.280546
365	X86	MINOR	1.2GHz	SINGLE_CHANNEL_DDR3_2133	0.06873	1.280169
366	X86	MINOR	1.2GHz	SINGLE_CHANNEL_DDR4_2400	0.068734	1.280254
367	X86	MINOR	1.2GHz	SINGLE_CHANNEL_DDR3_1600	0.068734	1.280252
368	X86	MINOR	1.2GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.068759	1.280705
369	X86	MINOR	1.2GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.068757	1.280677
370	X86	MINOR	1.2GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.068754	1.280621
371	X86	MINOR	1.2GHz	DUAL_CHANNEL_DDR3_1600	0.068733	1.280232
372	X86	MINOR	1.2GHz	DUAL_CHANNEL_DDR3_2133	0.068729	1.28016
373	X86	MINOR	1.2GHz	DUAL_CHANNEL_LPDDR3_1600	0.068747	1.280486
374	X86	MINOR	1.2GHz	DUAL_CHANNEL_DDR4_2400	0.068735	1.280256
375	X86	MINOR	1.4GHz	SINGLE_CHANNEL_DDR3_1600	0.058923	1.28043
376	X86	MINOR	1.4GHz	SINGLE_CHANNEL_DDR3_2133	0.058919	1.280331
377	X86	MINOR	1.4GHz	SINGLE_CHANNEL_LPDDR3_1600	0.058938	1.280746
378	X86	MINOR	1.4GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.058948	1.280956
379	X86	MINOR	1.4GHz	SINGLE_CHANNEL_DDR4_2400	0.058923	1.280417
380	X86	MINOR	1.4GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.058944	1.280889
381	X86	MINOR	1.4GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.058944	1.280868
382	X86	MINOR	1.4GHz	DUAL_CHANNEL_DDR3_1600	0.058922	1.280406
383	X86	MINOR	1.4GHz	DUAL_CHANNEL_DDR3_2133	0.058919	1.280339
384	X86	MINOR	1.4GHz	DUAL_CHANNEL_LPDDR3_1600	0.058936	1.280715
385	X86	MINOR	1.4GHz	DUAL_CHANNEL_DDR4_2400	0.058922	1.280406
386	X86	MINOR	1.6GHz	SINGLE_CHANNEL_DDR3_2133	0.051582	1.28051
387	X86	MINOR	1.6GHz	SINGLE_CHANNEL_DDR3_1600	0.051585	1.280599
388	X86	MINOR	1.6GHz	SINGLE_CHANNEL_LPDDR3_1600	0.051601	1.280981
389	X86	MINOR	1.6GHz	SINGLE_CHANNEL_DDR4_2400	0.051588	1.280666
390	X86	MINOR	1.6GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.05161	1.28122
391	X86	MINOR	1.6GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.051608	1.281165

392	X86	MINOR	1.6GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.051607	1.281127
393	X86	MINOR	1.6GHz	DUAL_CHANNEL_DDR3_1600	0.051584	1.280574
394	X86	MINOR	1.6GHz	DUAL_CHANNEL_DDR3_2133	0.051582	1.280516
395	X86	MINOR	1.6GHz	DUAL_CHANNEL_LPDDR3_1600	0.051599	1.280946
396	X86	MINOR	1.8GHz	SINGLE_CHANNEL_DDR3_1600	0.045896	1.280766
397	X86	MINOR	1.8GHz	SINGLE_CHANNEL_LPDDR3_1600	0.045912	1.281196
398	X86	MINOR	1.8GHz	SINGLE_CHANNEL_DDR3_2133	0.045894	1.280708
399	X86	MINOR	1.8GHz	SINGLE_CHANNEL_DDR4_2400	0.045897	1.280773
400	X86	MINOR	1.8GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.045921	1.281445
401	X86	MINOR	1.8GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.045918	1.281377
402	X86	MINOR	1.8GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.045918	1.281377
403	X86	MINOR	1.8GHz	DUAL_CHANNEL_DDR3_1600	0.045897	1.280773
404	X86	MINOR	1.8GHz	DUAL_CHANNEL_DDR3_2133	0.045894	1.280697
405	X86	MINOR	1.8GHz	DUAL_CHANNEL_LPDDR3_1600	0.045912	1.281193
406	X86	MINOR	1.8GHz	DUAL_CHANNEL_DDR4_2400	0.045896	1.280769
407	X86	MINOR	2GHz	SINGLE_CHANNEL_DDR3_2133	0.041278	1.280884
408	X86	MINOR	2GHz	SINGLE_CHANNEL_DDR3_1600	0.041281	1.280983
409	X86	MINOR	2GHz	SINGLE_CHANNEL_LPDDR3_1600	0.041296	1.281451
410	X86	MINOR	2GHz	SINGLE_CHANNEL_DDR4_2400	0.04128	1.28097
411	X86	MINOR	2GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.041305	1.28173
412	X86	MINOR	2GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.041302	1.281631
413	X86	MINOR	2GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.041301	1.2816
414	X86	MINOR	2GHz	DUAL_CHANNEL_DDR3_1600	0.04128	1.280955
415	X86	MINOR	2GHz	DUAL_CHANNEL_DDR3_2133	0.041277	1.280865
416	X86	MINOR	2GHz	DUAL_CHANNEL_LPDDR3_1600	0.041294	1.281401
417	X86	MINOR	2GHz	DUAL_CHANNEL_DDR4_2400	0.04128	1.280968
418	X86	MINOR	1.6GHz	DUAL_CHANNEL_DDR4_2400	0.051586	1.280615
419	X86	MINOR	2.2GHz	SINGLE_CHANNEL_DDR3_1600	0.037572	1.281197
420	X86	MINOR	2.2GHz	SINGLE_CHANNEL_LPDDR3_1600	0.037585	1.281667
421	X86	MINOR	2.2GHz	SINGLE_CHANNEL_DDR3_2133	0.037567	1.281028
422	X86	MINOR	2.2GHz	SINGLE_CHANNEL_DDR4_2400	0.037571	1.281173
423	X86	MINOR	2.2GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.037595	1.281978
424	X86	MINOR	2.2GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.037593	1.28193
425	X86	MINOR	2.2GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.037592	1.281873
426	X86	MINOR	2.2GHz	DUAL_CHANNEL_DDR3_1600	0.03757	1.28114
427	X86	MINOR	2.2GHz	DUAL_CHANNEL_DDR3_2133	0.037568	1.281079
428	X86	MINOR	2.2GHz	DUAL_CHANNEL_LPDDR3_1600	0.037585	1.281647
429	X86	MINOR	2.2GHz	DUAL_CHANNEL_DDR4_2400	0.037571	1.281171
430	X86	MINOR	2.4GHz	SINGLE_CHANNEL_LPDDR3_1600	0.034454	1.281943
431	X86	MINOR	2.4GHz	SINGLE_CHANNEL_DDR3_1600	0.034438	1.281347
432	X86	MINOR	2.4GHz	SINGLE_CHANNEL_DDR3_2133	0.034436	1.28126
433	X86	MINOR	2.4GHz	SINGLE_CHANNEL_DDR4_2400	0.034438	1.281352
434	X86	MINOR	2.4GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.034462	1.282227
435	X86	MINOR	2.4GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.03446	1.282162

436	X86	MINOR	2.4GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.03446	1.28216
437	X86	MINOR	2.4GHz	DUAL_CHANNEL_DDR3_1600	0.034438	1.281334
438	X86	MINOR	2.4GHz	DUAL_CHANNEL_DDR3_2133	0.034435	1.281242
439	X86	MINOR	2.4GHz	DUAL_CHANNEL_LPDDR3_1600	0.034452	1.281885
440	X86	MINOR	2.4GHz	DUAL_CHANNEL_DDR4_2400	0.034438	1.281355
441	X86	MINOR	2.6GHz	SINGLE_CHANNEL_DDR3_1600	0.0318	1.281552
442	X86	MINOR	2.6GHz	SINGLE_CHANNEL_DDR3_2133	0.031797	1.281406
443	X86	MINOR	2.6GHz	SINGLE_CHANNEL_DDR4_2400	0.0318	1.281551
444	X86	MINOR	2.6GHz	SINGLE_CHANNEL_LPDDR3_1600	0.031815	1.282154
445	X86	MINOR	2.6GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.031824	1.282523
446	X86	MINOR	2.6GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.031821	1.282405
447	X86	MINOR	2.6GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.03182	1.282367
448	X86	MINOR	2.6GHz	DUAL_CHANNEL_DDR3_1600	0.0318	1.281543
449	X86	MINOR	2.6GHz	DUAL_CHANNEL_DDR3_2133	0.031796	1.281386
450	X86	MINOR	2.6GHz	DUAL_CHANNEL_LPDDR3_1600	0.031814	1.28212
451	X86	MINOR	2.6GHz	DUAL_CHANNEL_DDR4_2400	0.031799	1.281506
452	X86	MINOR	2.8GHz	SINGLE_CHANNEL_DDR3_1600	0.029492	1.281754
453	X86	MINOR	2.8GHz	SINGLE_CHANNEL_DDR3_2133	0.029489	1.281605
454	X86	MINOR	2.8GHz	SINGLE_CHANNEL_LPDDR3_1600	0.029508	1.282426
455	X86	MINOR	2.8GHz	SINGLE_CHANNEL_DDR4_2400	0.029492	1.281727
456	X86	MINOR	2.8GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.029516	1.282785
457	X86	MINOR	2.8GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.029514	1.282714
458	X86	MINOR	2.8GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.029514	1.282691
459	X86	MINOR	2.8GHz	DUAL_CHANNEL_DDR3_1600	0.029492	1.28174
460	X86	MINOR	2.8GHz	DUAL_CHANNEL_DDR3_2133	0.029488	1.281593
461	X86	MINOR	2.8GHz	DUAL_CHANNEL_LPDDR3_1600	0.029506	1.282376
462	X86	MINOR	2.8GHz	DUAL_CHANNEL_DDR4_2400	0.029491	1.281708
463	X86	MINOR	3GHz	SINGLE_CHANNEL_DDR3_2133	0.027509	1.281741
464	X86	MINOR	3GHz	SINGLE_CHANNEL_DDR3_1600	0.027513	1.281939
465	X86	MINOR	3GHz	SINGLE_CHANNEL_LPDDR3_1600	0.027529	1.282661
466	X86	MINOR	3GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.027539	1.283109
467	X86	MINOR	3GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.027535	1.282923
468	X86	MINOR	3GHz	SINGLE_CHANNEL_DDR4_2400	0.027513	1.281919
469	X86	MINOR	3GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.027535	1.282924
470	X86	MINOR	3GHz	DUAL_CHANNEL_DDR3_1600	0.027512	1.281868
471	X86	MINOR	3GHz	DUAL_CHANNEL_DDR3_2133	0.027509	1.281731
472	X86	MINOR	3GHz	DUAL_CHANNEL_LPDDR3_1600	0.027528	1.282611
473	X86	MINOR	3GHz	DUAL_CHANNEL_DDR4_2400	0.027512	1.281863
474	X86	MINOR	3.2GHz	SINGLE_CHANNEL_DDR3_1600	0.025864	1.282097
475	X86	MINOR	3.2GHz	SINGLE_CHANNEL_DDR3_2133	0.025861	1.281952
476	X86	MINOR	3.2GHz	SINGLE_CHANNEL_LPDDR3_1600	0.02588	1.282879
477	X86	MINOR	3.2GHz	SINGLE_CHANNEL_DDR4_2400	0.025864	1.282078
478	X86	MINOR	3.2GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.025888	1.283257
479	X86	MINOR	3.2GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.025887	1.283219

480	X86	MINOR	3.2GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.025886	1.283161
481	X86	MINOR	3.2GHz	DUAL_CHANNEL_DDR3_1600	0.025864	1.282081
482	X86	MINOR	3.2GHz	DUAL_CHANNEL_DDR3_2133	0.02586	1.281902
483	X86	MINOR	3.2GHz	DUAL_CHANNEL_LPDDR3_1600	0.025878	1.282783
484	X86	MINOR	3.2GHz	DUAL_CHANNEL_DDR4_2400	0.025862	1.282011
485	X86	MINOR	3.3GHz	SINGLE_CHANNEL_DDR3_1600	0.02504	1.282197
486	X86	MINOR	3.3GHz	SINGLE_CHANNEL_LPDDR3_1600	0.025056	1.283009
487	X86	MINOR	3.3GHz	SINGLE_CHANNEL_DDR4_2400	0.025039	1.282167
488	X86	MINOR	3.3GHz	SINGLE_CHANNEL_DDR3_2133	0.025037	1.282031
489	X86	MINOR	3.3GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.025064	1.283417
490	X86	MINOR	3.3GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.025061	1.283295
491	X86	MINOR	3.3GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.025062	1.283328
492	X86	MINOR	3.3GHz	DUAL_CHANNEL_DDR3_1600	0.025039	1.282167
493	X86	MINOR	3.3GHz	DUAL_CHANNEL_DDR3_2133	0.025037	1.282031
494	X86	MINOR	3.3GHz	DUAL_CHANNEL_LPDDR3_1600	0.025054	1.282901
495	X86	MINOR	3.3GHz	DUAL_CHANNEL_DDR4_2400	0.025039	1.282137
496	X86	O3	600MHz	SINGLE_CHANNEL_DDR3_1600	0.026573	0.247327
497	X86	O3	600MHz	SINGLE_CHANNEL_DDR3_2133	0.026572	0.247314
498	X86	O3	600MHz	SINGLE_CHANNEL_LPDDR3_1600	0.026583	0.247417
499	X86	O3	600MHz	SINGLE_CHANNEL_DDR4_2400	0.026572	0.247314
500	X86	O3	600MHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.026586	0.247448
501	X86	O3	600MHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.026586	0.247449
502	X86	O3	600MHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.026586	0.247449
503	X86	O3	600MHz	DUAL_CHANNEL_DDR3_1600	0.026572	0.247319
504	X86	O3	600MHz	DUAL_CHANNEL_DDR3_2133	0.026572	0.247314
505	X86	O3	600MHz	DUAL_CHANNEL_LPDDR3_1600	0.02658	0.247391
506	X86	O3	600MHz	DUAL_CHANNEL_DDR4_2400	0.026572	0.247314
507	X86	O3	800MHz	SINGLE_CHANNEL_DDR3_1600	0.019932	0.247401
508	X86	O3	800MHz	SINGLE_CHANNEL_DDR3_2133	0.019931	0.247387
509	X86	O3	800MHz	SINGLE_CHANNEL_LPDDR3_1600	0.019941	0.247519
510	X86	O3	800MHz	SINGLE_CHANNEL_DDR4_2400	0.019931	0.247396
511	X86	O3	800MHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.019946	0.247578
512	X86	O3	800MHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.019944	0.247551
513	X86	O3	800MHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.019944	0.247554
514	X86	O3	800MHz	DUAL_CHANNEL_DDR3_1600	0.019931	0.247391
515	X86	O3	800MHz	DUAL_CHANNEL_DDR3_2133	0.01993	0.247381
516	X86	O3	800MHz	DUAL_CHANNEL_LPDDR3_1600	0.01994	0.247499
517	X86	O3	800MHz	DUAL_CHANNEL_DDR4_2400	0.019931	0.24739
518	X86	O3	1GHz	SINGLE_CHANNEL_DDR3_1600	0.015951	0.247495
519	X86	O3	1GHz	SINGLE_CHANNEL_DDR3_2133	0.01595	0.247469
520	X86	O3	1GHz	SINGLE_CHANNEL_LPDDR3_1600	0.01596	0.247635
521	X86	O3	1GHz	SINGLE_CHANNEL_DDR4_2400	0.015955	0.247555
522	X86	O3	1GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.015962	0.247655
523	X86	O3	1GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.015961	0.247639

524	X86	O3	1GHz	DUAL_CHANNEL_DDR3_1600	0.01595	0.247476
525	X86	O3	1GHz	DUAL_CHANNEL_DDR3_2133	0.015949	0.247453
526	X86	O3	1GHz	DUAL_CHANNEL_LPDDR3_1600	0.015959	0.247607
527	X86	O3	1GHz	DUAL_CHANNEL_DDR4_2400	0.015951	0.247487
528	X86	O3	1.2GHz	SINGLE_CHANNEL_DDR3_1600	0.013292	0.247583
529	X86	O3	1.2GHz	SINGLE_CHANNEL_LPDDR3_1600	0.013302	0.24776
530	X86	O3	1.2GHz	SINGLE_CHANNEL_DDR3_2133	0.01329	0.247541
531	X86	O3	1.2GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.013303	0.247783
532	X86	O3	1.2GHz	SINGLE_CHANNEL_DDR4_2400	0.013292	0.247578
533	X86	O3	1.2GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.013302	0.247771
534	X86	O3	1.2GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.013302	0.247769
535	X86	O3	1.2GHz	DUAL_CHANNEL_DDR3_1600	0.013291	0.24756
536	X86	O3	1.2GHz	DUAL_CHANNEL_DDR3_2133	0.01329	0.24754
537	X86	O3	1.2GHz	DUAL_CHANNEL_LPDDR3_1600	0.0133	0.24773
538	X86	O3	1.2GHz	DUAL_CHANNEL_DDR4_2400	0.013292	0.247578
539	X86	O3	1GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.015964	0.247684
540	X86	O3	1.4GHz	SINGLE_CHANNEL_DDR3_2133	0.011396	0.24763
541	X86	O3	1.4GHz	SINGLE_CHANNEL_LPDDR3_1600	0.011407	0.247889
542	X86	O3	1.4GHz	SINGLE_CHANNEL_DDR3_1600	0.011398	0.247677
543	X86	O3	1.4GHz	SINGLE_CHANNEL_DDR4_2400	0.011396	0.247649
544	X86	O3	1.4GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.011409	0.247928
545	X86	O3	1.4GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.011409	0.247916
546	X86	O3	1.4GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.011407	0.247883
547	X86	O3	1.4GHz	DUAL_CHANNEL_DDR3_1600	0.011396	0.247651
548	X86	O3	1.4GHz	DUAL_CHANNEL_DDR3_2133	0.011395	0.247617
549	X86	O3	1.4GHz	DUAL_CHANNEL_LPDDR3_1600	0.011406	0.247853
550	X86	O3	1.4GHz	DUAL_CHANNEL_DDR4_2400	0.011396	0.247644
551	X86	O3	1.6GHz	SINGLE_CHANNEL_DDR3_1600	0.00998	0.24775
552	X86	O3	1.6GHz	SINGLE_CHANNEL_DDR3_2133	0.009979	0.247718
553	X86	O3	1.6GHz	SINGLE_CHANNEL_DDR4_2400	0.00998	0.247745
554	X86	O3	1.6GHz	SINGLE_CHANNEL_LPDDR3_1600	0.00999	0.248008
555	X86	O3	1.6GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.009992	0.248051
556	X86	O3	1.6GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.009991	0.248018
557	X86	O3	1.6GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.00999	0.248
558	X86	O3	1.6GHz	DUAL_CHANNEL_DDR3_1600	0.009979	0.247722
559	X86	O3	1.6GHz	DUAL_CHANNEL_DDR3_2133	0.009978	0.247694
560	X86	O3	1.6GHz	DUAL_CHANNEL_LPDDR3_1600	0.009988	0.247941
561	X86	O3	1.6GHz	DUAL_CHANNEL_DDR4_2400	0.00998	0.247747
562	X86	O3	1.8GHz	SINGLE_CHANNEL_DDR3_1600	0.008882	0.247853
563	X86	O3	1.8GHz	SINGLE_CHANNEL_DDR3_2133	0.00888	0.24779
564	X86	O3	1.8GHz	SINGLE_CHANNEL_LPDDR3_1600	0.008891	0.248114
565	X86	O3	1.8GHz	SINGLE_CHANNEL_DDR4_2400	0.008882	0.247861
566	X86	O3	1.8GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.008893	0.248175
567	X86	O3	1.8GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.008892	0.248131

568	X86	O3	1.8GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.008892	0.248123
569	X86	O3	1.8GHz	DUAL_CHANNEL_DDR3_1600	0.00888	0.247806
570	X86	O3	1.8GHz	DUAL_CHANNEL_DDR3_2133	0.008879	0.247779
571	X86	O3	1.8GHz	DUAL_CHANNEL_LPDDR3_1600	0.00889	0.24807
572	X86	O3	1.8GHz	DUAL_CHANNEL_DDR4_2400	0.00888	0.247793
573	X86	O3	2GHz	SINGLE_CHANNEL_DDR3_2133	0.007989	0.247914
574	X86	O3	2GHz	SINGLE_CHANNEL_LPDDR3_1600	0.008	0.248236
575	X86	O3	2GHz	SINGLE_CHANNEL_DDR3_1600	0.007991	0.247966
576	X86	O3	2GHz	SINGLE_CHANNEL_DDR4_2400	0.007991	0.247963
577	X86	O3	2GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.008001	0.248293
578	X86	O3	2GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.008001	0.248291
579	X86	O3	2GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.008	0.248238
580	X86	O3	2GHz	DUAL_CHANNEL_DDR3_1600	0.007991	0.247969
581	X86	O3	2GHz	DUAL_CHANNEL_DDR3_2133	0.007987	0.247837
582	X86	O3	2GHz	DUAL_CHANNEL_LPDDR3_1600	0.007997	0.24817
583	X86	O3	2GHz	DUAL_CHANNEL_DDR4_2400	0.007987	0.247843
584	X86	O3	2.2GHz	SINGLE_CHANNEL_DDR3_1600	0.007274	0.248027
585	X86	O3	2.2GHz	SINGLE_CHANNEL_DDR3_2133	0.007271	0.24794
586	X86	O3	2.2GHz	SINGLE_CHANNEL_LPDDR3_1600	0.007284	0.24837
587	X86	O3	2.2GHz	SINGLE_CHANNEL_DDR4_2400	0.007273	0.248001
588	X86	O3	2.2GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.007286	0.24846
589	X86	O3	2.2GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.007284	0.248401
590	X86	O3	2.2GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.007284	0.2484
591	X86	O3	2.2GHz	DUAL_CHANNEL_DDR3_1600	0.007272	0.247969
592	X86	O3	2.2GHz	DUAL_CHANNEL_DDR3_2133	0.00727	0.247924
593	X86	O3	2.2GHz	DUAL_CHANNEL_LPDDR3_1600	0.007281	0.248289
594	X86	O3	2.2GHz	DUAL_CHANNEL_DDR4_2400	0.007271	0.24794
595	X86	O3	2.4GHz	SINGLE_CHANNEL_DDR3_1600	0.006669	0.248118
596	X86	O3	2.4GHz	SINGLE_CHANNEL_DDR3_2133	0.006667	0.248065
597	X86	O3	2.4GHz	SINGLE_CHANNEL_LPDDR3_1600	0.006679	0.248493
598	X86	O3	2.4GHz	SINGLE_CHANNEL_DDR4_2400	0.006668	0.248081
599	X86	O3	2.4GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.00668	0.248558
600	X86	O3	2.4GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.006679	0.248505
601	X86	O3	2.4GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.006679	0.248502
602	X86	O3	2.4GHz	DUAL_CHANNEL_DDR3_1600	0.006667	0.248069
603	X86	O3	2.4GHz	DUAL_CHANNEL_DDR3_2133	0.006666	0.248034
604	X86	O3	2.4GHz	DUAL_CHANNEL_LPDDR3_1600	0.006676	0.248397
605	X86	O3	2.4GHz	DUAL_CHANNEL_DDR4_2400	0.006666	0.24804
606	X86	O3	2.6GHz	SINGLE_CHANNEL_DDR3_1600	0.006159	0.248189
607	X86	O3	2.6GHz	SINGLE_CHANNEL_LPDDR3_1600	0.006169	0.248622
608	X86	O3	2.6GHz	SINGLE_CHANNEL_DDR3_2133	0.006157	0.24813
609	X86	O3	2.6GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.006171	0.248686
610	X86	O3	2.6GHz	SINGLE_CHANNEL_DDR4_2400	0.006158	0.248154
611	X86	O3	2.6GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.00617	0.248638

612	X86	O3	2.6GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.006169	0.248607
613	X86	O3	2.6GHz	DUAL_CHANNEL_DDR3_1600	0.006157	0.248134
614	X86	O3	2.6GHz	DUAL_CHANNEL_DDR3_2133	0.006156	0.248081
615	X86	O3	2.6GHz	DUAL_CHANNEL_LPDDR3_1600	0.006167	0.248539
616	X86	O3	2.6GHz	DUAL_CHANNEL_DDR4_2400	0.006156	0.248103
617	X86	O3	2.8GHz	SINGLE_CHANNEL_DDR3_2133	0.005711	0.248206
618	X86	O3	2.8GHz	SINGLE_CHANNEL_LPDDR3_1600	0.005724	0.248749
619	X86	O3	2.8GHz	SINGLE_CHANNEL_DDR3_1600	0.005713	0.24828
620	X86	O3	2.8GHz	SINGLE_CHANNEL_DDR4_2400	0.005712	0.248242
621	X86	O3	2.8GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.005725	0.248811
622	X86	O3	2.8GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.005725	0.248813
623	X86	O3	2.8GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.005725	0.248808
624	X86	O3	2.8GHz	DUAL_CHANNEL_DDR3_1600	0.005712	0.248229
625	X86	O3	2.8GHz	DUAL_CHANNEL_DDR3_2133	0.00571	0.248157
626	X86	O3	2.8GHz	DUAL_CHANNEL_LPDDR3_1600	0.005721	0.248656
627	X86	O3	2.8GHz	DUAL_CHANNEL_DDR4_2400	0.005711	0.24819
628	X86	O3	3GHz	SINGLE_CHANNEL_DDR3_1600	0.005331	0.248398
629	X86	O3	3GHz	SINGLE_CHANNEL_DDR3_2133	0.005329	0.248285
630	X86	O3	3GHz	SINGLE_CHANNEL_DDR4_2400	0.00533	0.248352
631	X86	O3	3GHz	SINGLE_CHANNEL_LPDDR3_1600	0.005341	0.248841
632	X86	O3	3GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.005343	0.248944
633	X86	O3	3GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.005342	0.248894
634	X86	O3	3GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.005342	0.248884
635	X86	O3	3GHz	DUAL_CHANNEL_DDR3_1600	0.00533	0.248356
636	X86	O3	3GHz	DUAL_CHANNEL_DDR3_2133	0.005327	0.248224
637	X86	O3	3GHz	DUAL_CHANNEL_LPDDR3_1600	0.005339	0.248747
638	X86	O3	3GHz	DUAL_CHANNEL_DDR4_2400	0.005328	0.248253
639	X86	O3	3.2GHz	SINGLE_CHANNEL_DDR3_2133	0.00501	0.248361
640	X86	O3	3.2GHz	SINGLE_CHANNEL_LPDDR3_1600	0.005022	0.248967
641	X86	O3	3.2GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.005025	0.249089
642	X86	O3	3.2GHz	SINGLE_CHANNEL_DDR4_2400	0.005011	0.248402
643	X86	O3	3.2GHz	SINGLE_CHANNEL_DDR3_1600	0.005012	0.248448
644	X86	O3	3.2GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.005023	0.248996
645	X86	O3	3.2GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.005024	0.249043
646	X86	O3	3.2GHz	DUAL_CHANNEL_DDR3_1600	0.005011	0.248376
647	X86	O3	3.2GHz	DUAL_CHANNEL_DDR3_2133	0.00501	0.248324
648	X86	O3	3.2GHz	DUAL_CHANNEL_LPDDR3_1600	0.005021	0.248868
649	X86	O3	3.2GHz	DUAL_CHANNEL_DDR4_2400	0.00501	0.24833
650	X86	O3	3.3GHz	SINGLE_CHANNEL_DDR3_2133	0.004851	0.248389
651	X86	O3	3.3GHz	SINGLE_CHANNEL_LPDDR3_1600	0.004864	0.249059
652	X86	O3	3.3GHz	SINGLE_CHANNEL_DDR4_2400	0.004853	0.248485
653	X86	O3	3.3GHz	SINGLE_CHANNEL_DIMM_DDR5_4400	0.004865	0.249135
654	X86	O3	3.3GHz	SINGLE_CHANNEL_DDR3_1600	0.004853	0.248526
655	X86	O3	3.3GHz	SINGLE_CHANNEL_DIMM_DDR5_6400	0.004864	0.249063

656	X86	O3	3.3GHz	SINGLE_CHANNEL_DIMM_DDR5_8400	0.004864	0.249063
657	X86	O3	3.3GHz	DUAL_CHANNEL_DDR3_1600	0.004852	0.248438
658	X86	O3	3.3GHz	DUAL_CHANNEL_DDR3_2133	0.00485	0.248342
659	X86	O3	3.3GHz	DUAL_CHANNEL_LPDDR3_1600	0.004861	0.248934
660	X86	O3	3.3GHz	DUAL_CHANNEL_DDR4_2400	0.004851	0.248411